The Equilibrium Effects of Revenue-Sharing Agreements*

Fernando Luco† Guillermo Marshall‡

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Abstract

Vertical contracts govern firms' incentives along the supply chain. Revenue-sharing agreements (RSAs) are a commonly used contract to satisfy incentive constraints but have a theoretically ambiguous impact on joint supply chain profits. On the one hand, RSAs generate an upward pricing pressure that leads to inefficiency and profit losses relative to vertical integration (all else equal). On the other hand, RSAs act as a commitment device to price less aggressively, which may lessen the intensity of competition. We leverage variation in vertical structure, institutional knowledge, and a novel methodology to estimate the parameters of a revenue-sharing agreement involving General Mills and an upstream firm in the U.S. yogurt industry. We find that General Mills remitted 3 to 5 percent of its Yoplait yogurt revenue to the upstream firm prior to vertical integration. Using these estimates in conjunction with an empirical model of supply and demand, we quantify the equilibrium impacts of RSAs relative to vertical integration. Eliminating the agreement lowered Yoplait retail prices by 1.9 to 3.1 percent and raised the downstream firm's profits by 9.4 to 15.7 percent. Eliminating the RSA increased joint supply-chain profits by 0.4 to 0.85 percent, which gives a lower bound on the value of incentive frictions along a decentralized supply chain.

Keywords: Boundaries of the firm, vertical contracts, revenue-sharing agreement, vertical integration

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†Texas A&M University, Department of Economics, email: fluco@tamu.edu

‡University of British Columbia, Sauder School of Business, email: guillermo.marshall@sauder.ubc.ca

1 Introduction

Firms face the challenge of designing contracts to align incentives along the vertical supply chain. Monitoring problems, asymmetries of information, or incentive issues more broadly may prevent vertically independent firms from replicating the outcomes of a vertically integrated supply chain. While revenue-sharing agreements are a common contract used by supply chains to tackle incentive issues, their equilibrium impact on supply chain profitability is theoretically ambiguous due to two competing economic forces.

On the one hand, a revenue-sharing agreement distorts pricing incentives, which may lead to inefficient prices and lower joint supply chain profits—see, e.g., Gil (2009) for evidence from the movie industry. To see why, consider an agreement where a downstream firm retains a fraction $\mu \in (0, 1)$ of its revenue, remitting the rest upstream. Even assuming the upstream firm transfers intermediate inputs at marginal cost (avoiding standard double marginalization), the downstream firm maximizes its private profit by setting $\mu \times \mathrm{MR} = \mathrm{MC}$. Unlike a vertically integrated supply chain that sets $\mathrm{MR} = \mathrm{MC}$, the independent downstream firm acts as if its marginal costs are inflated ($\mathrm{MR} = \mathrm{MC}/\mu > \mathrm{MC}$). This upward pricing pressure may induce inefficiency along the supply chain, decreasing the quantity sold and, thus, joint supply chain profits (all else equal).

On the other hand, in an oligopoly setting, this exact distortion can act as a profitable commitment device. By artificially inflating the downstream firm's "effective" marginal costs (MC/μ as opposed to MC), a revenue-sharing agreement commits the firm to less aggressive pricing. Due to strategic complementarities, rival firms are incentivized to respond by raising their own prices. As highlighted by the strategic delegation literature (e.g., Fershtman and Judd, 1987), this softening of competition can offset the efficiency loss of the contract and even increase the joint profits of the supply chain. Whether a revenue-sharing agreement increases or decreases joint supply chain profits is therefore an open empirical question.

In this paper, we address the question of the equilibrium impacts of a revenue-sharing agreement by exploiting variation in vertical structure—a supply chain's transition from a revenue-sharing agreement to vertical integration. Because the specific parameters of business-to-business contracts (such as μ in the example above) are typically confidential, we first develop a novel methodology to estimate the unobserved contract terms using a structural supply and demand model, where we leverage this ownership transition. We then quantify the contract's equilibrium impacts on prices, quantities, and the distribution of profits. In doing so, we contribute new evidence on the economics of vertical practices (see, e.g., Shaffer, 1991; Asker and Bar-Isaac, 2014), the economic value of incentive issues in a decentralized supply chain, and the impact of firm boundaries on market outcomes (see, e.g., Mullainathan and Scharfstein, 2001; Hortaçsu and Syverson, 2007; Lafontaine and Slade, 2007).

Our setting is the U.S. yogurt industry. The supply chain of top-selling Yoplait products in the U.S. has featured a stable structure over several decades. General Mills and Société de Diffusion de Marque (Sodima/Sodiaal, hereafter) signed a contract in 1977 that, though it has been amended several times, lasted until 2011.[1] Between 1977 and 2011, General Mills sold Yoplait in the U.S. under a licensing contract that featured a revenue-sharing agreement. General Mills then acquired the Yoplait brand in July 2011. Before the transaction, Sodima/Sodiaal was a supplier of intangible inputs (e.g., brand or image rights) to General Mills; after the transaction, the supply chain became vertically integrated, terminating the revenue-sharing agreement.

The U.S. yogurt industry is ideal for our analyses for two reasons. First, public documents describe the structure of the revenue-sharing contract between General Mills and Sodima/Sodiaal (though the exact contract terms are redacted). Second, the industry featured a change in vertical structure caused by General Mills' acquisition of a controlling interest in Yoplait in 2011, which we interpret as a termination of the revenue-sharing agreement. The 2011 acquisition directly affected General Mills' pricing incentives, while all other firms' pricing incentives were affected indirectly through equilibrium effects.

To quantify the impact of the revenue-sharing agreement on market efficiency, we use weekly scanner data from 47 metropolitan areas in the U.S. from the IRI Marketing Data Set (Bronnenberg et al., 2008). We focus on the largest three yogurt manufacturers in the U.S., covering 75.6% of the category revenues between 2010 and 2012. We complement these data with public documents that reveal the structure of the revenue-sharing agreement between General Mills and Sodima, as well as demographic information from the 2010 American Community Survey (Ruggles et al., 2025).

In our work, we follow a two-step research design. First, we exploit variation in vertical structure to measure the equilibrium price effects of revenue-sharing agreements. Specifically, we compare within-product price changes among products that were subject to the revenue-sharing agreement, with those of products that were not subject to the agreement (e.g., Dannon products), before and after the transaction. We show that prices of Yoplait products decreased by 1.8 to 3.8 percent after General Mills acquired the Yoplait brand, relative to the prices of products that were not subject to the agreement. Importantly, we show that the price changes took place soon after the transaction was finalized and that these price

[1] As we describe in Section 3.1, Sodima was restructured in 1989 and it became Sodiaal (for "Société de diffusion internationale agro-alimentaire"), and in 2002 Sodiaal established a 50/50 partnership with PAI Partners, a private equity firm. Across all these years, Yoplait remained owned by Sodima, Sodiaal, and the Sodiaal-PAI partnership.

reductions lasted through the end of our sample period.

Second, after establishing that the shift from a revenue-sharing contract to vertical integration impacted the prices of Yoplait products, as the theory predicts, we turn to quantifying the equilibrium impacts of the contract. We do this in three steps. First, we estimate demand for yogurt following the standard approach in the literature (Conlon and Gortmaker, 2020), including work in the U.S. yogurt industry (Villas-Boas, 2007; Hristakeva, 2022; Duarte et al., 2024b, among others). We then use the demand estimates, together with the first-order conditions of the pricing problem of yogurt manufacturers, to recover the firms' marginal costs and estimate a key parameter governing the revenue-sharing contract between General Mills and Sodima/Sodiaal.

We estimate this revenue-sharing agreement parameter using two complementary approaches. In the first one, we use the demand estimates to recover the ratio of marginal costs and the revenue-sharing parameter (i.e., the marginal cost mc_{jt} for products of non-Yoplait manufacturers, and the ratio mc_{jt}/μ for Yoplait products before July 2011, and mc_{jt} afterward). We then use this ratio as the dependent variable in a linear regression that allows us to identify μ under the assumption that there were no systematic changes in the marginal costs of Yoplait products around the time of the transaction. We estimate that Sodima received about 3 percent of General Mills' Yoplait revenues. As we mentioned above, and show in Section 4, because there were no changes in relative prices and in price levels prior to the transaction, we are reassured that these estimates were caused by the transition from revenue sharing to vertical integration.

In our second strategy, we use an indirect inference approach to identify the revenue-sharing parameter. We start by making an arbitrary guess for this parameter, which directly impacts our marginal cost estimates, as our demand estimates and supply model allow us to recover marginal costs up to the revenue-sharing parameter (see previous paragraph). Next, using our demand and supply model, we simulate what equilibrium market prices would be if no revenue-sharing agreement existed. With these simulated prices in hand, we compare how Yoplait's prices shifted relative to competitors before and after July 2011. Assuming the contract's termination was the only major industry change during this time, evaluating the model at the true revenue-sharing value should completely strip away any price distortions—resulting in a prediction of zero relative price change. Therefore, we test different guesses until we find the exact value that produces this null effect. Using this approach, we estimate that Sodima received about 3 to 5 percent of General Mills' Yoplait revenues (i.e., these are the values that induce a null effect in different subsamples).

With these estimates of the revenue-sharing parameter in hand, we turn to quantify the impact of the revenue-sharing contract on the entire market. To do this, we first compute market outcomes when the revenue-sharing contract was in place. Then, we compare these outcomes with the market outcomes under vertical integration (i.e., when the revenue-sharing contract is no longer in effect).

Replacing the revenue-sharing contract with vertical integration led to price decreases for Yoplait products of between 1.9 to 3.1 percent, depending on which estimate of the revenue-sharing parameter we consider. The price decreases in Yoplait products led to an increase in Yoplait's market share of between 6.2 and 10.3 percent.

We find that the RSA had a negative impact on the joint profits of the Yoplait supply chain of 0.4 to 0.85 percent. These estimates suggest that the inefficiency effect dominated the competition intensity effect of the RSA, making the RSA a surplus-decreasing contract. We also find that General Mills' profits increased by 9.4 to 15.7 percent when eliminating the revenue-sharing agreement, reflecting gains from efficient pricing and the elimination of revenue sharing with Sodiaal.

Our estimates imply that General Mills' annual profits increased by between \$62 million and \$104 million after eliminating the revenue-sharing agreement. Overall, our findings show that vertical integration led to lower prices, higher quantities, and higher profits, showing that the nature of vertical contracts can significantly impact supply chains and market outcomes.

Finally, we use these estimates to speak to why firms adopt RSAs despite their costs. The 0.4–0.85 percent surplus loss provides a lower bound on the value of the incentive frictions the RSA addressed. Its small magnitude, together with the cost of renegotiation, helps rationalize why the contract persisted largely unchanged for over three decades, until vertical integration eliminated the distortion at a transaction price of \$1.1 billion.

Literature Review

Given the impact of revenue-sharing agreements on pricing incentives and profits, a natural question is, 'Why do firms use them?' The literature has identified benefits to revenue-sharing agreements that may outweigh the above-mentioned costs.

Gallini and Wright (1990) consider the problem of a seller transferring a technology to a buyer in the presence of asymmetric information and the possibility of imitation. The uninformed buyer may refuse to make relationship-specific investments if not reassured about the value of the technology.[2] The authors show that in equilibrium, the informed seller will use a contract with output-related royalty to signal the value of a technology, which is not possible using only lump-sum payments. Beggs (1992) makes a similar point. Revenue-

[2] Gil and Lafontaine (2012) make a similar point in the context of movie exhibition contracts.

sharing agreements (where compensation increases with output) can also be profitable to use as a tool to incentivize costly effort (Bhattacharyya and Lafontaine, 1995; Lazear, 2000).

Mortimer (2008) considers the vertical relationship between the movie distributor and the downstream video rental firms, where the latter purchase video cassettes from the former. In particular, the author considers a transition from linear contracts (a lump sum payment for every cassette) to revenue-sharing agreements, in which the rental revenue of a video cassette was shared between the movie distributor and the downstream video rental firm. The author shows that a revenue-sharing agreement can improve the profits of the supply chain, as it leads to more efficient inventory choices. Unlike ours, Mortimer (2008) considers a setting where the revenue-sharing agreement does not distort output decisions (for a given inventory choice).

As discussed, profit-sharing and revenue-sharing agreements can signal the value of a technology, incentivize effort, or coordinate investments in some cases. A key difference between profit-sharing and revenue-sharing agreements, however, is that the latter distorts pricing incentives, which impacts the profits of the supply chain. Although pricing incentives are not distorted when using a profit-sharing agreement, economic profits are harder to monitor than revenue. When monitoring costs are greater than the profit loss due to distorted pricing incentives, a revenue-sharing agreement will be preferred.

Rather than analyzing the reasons for using a revenue-sharing agreement in the U.S. yogurt industry, our contribution lies in uncovering the details of a revenue-sharing agreement and quantifying its impact on market outcomes. More broadly, we contribute to the literature studying the economics of vertical mergers and vertical relationships.[3]

Our work also relates to the literature on strategic delegation (Spencer and Brander, 1983; Vickers, 1985; Fershtman and Judd, 1987; Sklivas, 1987; Bonanno and Vickers, 1988), which studies the problem of the firm owner in designing a managerial contract for the manager who chooses prices or quantities in an oligopoly game. While our focus is on a supply chain rather than a decentralized firm, similar insights arise in both cases. For example, Fershtman and Judd (1987) show that a firm owner may choose a managerial contract that distorts managers' incentives away from profit maximization so as to relax the intensity of product market competition. As mentioned above, revenue-sharing agreements induce a similar result in a supply chain, as these distort the downstream firm's pricing incentive in ways that may lead to less aggressive pricing. Our contribution is to empirically quantify the impact of an incentive distortion of this type (i.e., a revenue-sharing agreement) on supply

[3]See, for example, Villas-Boas (2007); Hortaçsu and Syverson (2007); Gil (2009); Bonnet and Dubois (2010); Crawford and Yurukoglu (2012); Houde (2012); Asker (2016); Crawford et al. (2018); Luco and Marshall (2020); Chen et al. (2024); Gil et al. (2024).

chain outcomes.

2 The Impacts of Revenue-Sharing Agreements

To examine the impact of a revenue-sharing agreement on supply chain outcomes, consider the following Hotelling-style example (Hotelling, 1929). Two downstream firms compete with each other selling one product each: firm A and firm B. Firm A's product requires an input product supplied by an upstream firm (for simplicity, firm B's product does not). Firm A compensates the upstream firm by paying a fraction $1-\mu$ of its downstream revenue, where $\mu \in (0, 1]$.[4] We assume that the marginal cost of the input product is zero, which, for example, may capture that the input is intangible (e.g., brand or image rights). The marginal cost of production of firms A and B is given by c.

The products of firm A and B are horizontally differentiated, and a consumer with a preference parameter x will choose firm A's product if and only if:

$$v + \delta - p_A - \tau \cdot x \geq v - p_B - \tau \cdot (1 - x),$$

where v, δ, and τ are preference parameters, δ captures the asymmetry between products, and p_A and p_B are the prices of firms A and B, respectively. We assume x is uniformly distributed on the unit interval.[5] The demand for firm A and firm B's products are given by $x(p_A, p_B)$ and $1 - x(p_A, p_B)$, respectively, where

$$x(p_A, p_B) = \frac{1}{2} + \frac{\delta + p_B - p_A}{2\tau}.$$

The firms simultaneously choose their prices by maximizing $\pi_j(p_A, p_B)$. Firm A's problem is given by $\max_{p_A} \pi_A(p_A, p_B) = (\mu \cdot p_A - c)x(p_A, p_B)$, where μ is the coefficient of the revenue-sharing agreement.

How does the revenue-sharing agreement impact profits and pricing incentives? On the one hand, the revenue-sharing agreement creates inefficiency. To see this, divide $\pi_A(p_A, p_B)$ by μ, and the objective function becomes $\tilde{\pi}_A(p_A, p_B) = (p_A - c/\mu)x(p_A, p_B)$. One can immediately notice that a revenue-sharing agreement (i.e., $\mu < 1$) is equivalent, from an incentive perspective, to scaling up the marginal cost by $1/\mu$, which creates upward pressure on prices that induces inefficiency and distorts profits.

The profit of firm A's supply chain is given by $\pi_A^{SC}(p_A, p_B; \mu) = (p_A - c)x(p_A, p_B)$ since the production cost of the input is zero and the upstream and downstream firms share

[4] We abstract away from lump-sum payments, as they do not impact marginal pricing incentives.

[5] We assume v is sufficiently large that every consumer is served by one of the firms.

the revenue. Whenever $\mu < 1$, the downstream firm chooses $p_A^*(\mu, p_B) = \arg\max(p_A - c/\mu)x(p_A, p_B)$, which does not equal the price that solves $\max_{p_A}(p_A - c)x(p_A, p_B)$. That is, the revenue-sharing agreement decreases the profit of the entire supply chain relative to what it would earn without a revenue-sharing agreement, holding the prices of the rival fixed.[6]

On the other hand, the revenue-sharing agreement relaxes the intensity of price competition. The Nash equilibrium of the game features

$$p_A^* = \tau + \frac{\delta}{3} + \frac{2}{3}\frac{c}{\mu} + \frac{c}{3} \quad \text{and} \quad x^* = \frac{1}{2} + \frac{\delta - c/\mu + c}{6\tau},$$

which shows that the price of firm A decreases in μ.[7] Note that the effect of the revenue-sharing agreement on profits is different than that of a higher marginal cost, as the revenue-sharing agreement changes pricing incentives (of all downstream firms) but does not impact the actual marginal cost of production of firm A.[8]

Using these equilibrium values, we can compute the profit of the supply chain, $\pi_A^{*,SC} = (p_A^* - c)x^*$, and derive the impact of the revenue-sharing agreement on the profits of the supply chain:

$$\frac{\partial \pi_A^{*,SC}}{\partial \mu} = \frac{c}{\mu^2 6\tau}\left(-\tau - \frac{\delta}{3} + \frac{4}{3}\frac{c}{\mu} - \frac{4}{3}c\right),$$

which can be positive or negative. That is, a revenue-sharing agreement may increase supply chain profits despite causing inefficiency.

This derivative reveals the conditions under which a revenue-sharing agreement is likelier to increase joint supply chain profits. For instance, consider the role of the product differentiation parameter, τ, which dictates the price elasticity of demand (i.e., a higher τ implies a more inelastic demand). As τ increases, the $-\tau$ term inside the parentheses becomes more dominant, making the derivative negative, which implies that joint supply chain profits increase as μ falls below 1. Economically, this implies that a highly differentiated downstream firm can afford to distort prices upward to soften competition without losing substantial market share to rivals. We show similar results in Online Appendix E in the context of an oligopoly model with a nested logit demand, where a revenue-sharing agreement may or may

[6] Cachon and Lariviere (2005) argue that a revenue-sharing agreement where the downstream firm pays the upstream firm a particular wholesale price for each unit sold, w, in addition to a fraction $1-\mu$ the revenue can lead to supply chain coordination (i.e., outcomes that equal those of a vertically-integrated supply chain). Setting $w = \mu c - c < 0$ in the example above would imply that the profit function of the downstream firm equals $\pi(p;\mu) = \mu \cdot p \cdot q(p) - (c + w) \cdot q(p) = \mu \cdot (p - c) \cdot q(p)$, effectively converting the revenue-sharing agreement into a profit-sharing agreement, aligning pricing incentives along the supply chain. The use of a wholesale price along these lines, however, may not be feasible in the Yoplait setting, where the transferred input is intangible.

[7] The equilibrium price of product B is given by $p_B^* = \tau - \delta/3 + 2c/3 + c/(3\mu)$.

[8] This is, while firm A operates as if its marginal cost is c/μ, the effective marginal cost is still c.

not increase joint supply chain profits depending on the precise competitive environment.

In summary, a revenue-sharing agreement distorts pricing incentives, causing inefficiency, but may also lessen the intensity of downstream price competition. The equilibrium impact of a revenue-sharing agreement on joint supply chain profits is ambiguous and thus an empirical question, which we turn to next.

3 Industry and Data

3.1 Industry Overview

In 1964, in France, farmers of various co-ops formed the "Société de Diffusion de Marque" (Sodima, hereafter). In 1965, Sodima combined two of its brands to form Yoplait. In 1974, Michigan Cottage Cheese Co. acquired the rights to produce and market Yoplait in the United States. Yoplait was launched in the United States in 1976 (General Mills, 2015).

General Mills started exploring the possibility of producing yogurt in 1975. In 1977, upon discovering Yoplait in the United States, General Mills acquired the production and marketing rights for the United States, and it acquired the production facility built by Michigan Cottage Cheese Co. in Reed City, Michigan (General Mills, 2015).

In 1989, Sodima became Sodiaal (the acronym for "Société de diffusion internationale agro-alimentaire"). In 2002, PAI Partners (a French private equity firm) and Sodiaal established a 50/50 partnership based on the Yoplait brand.

The contractual relationship between General Mills and Sodima/Sodiaal The 1977 license agreement between Sodima and General Mills granted General Mills a license to manufacture, distribute, and sell Yoplait in the United States.

The agreement had two main financial components.[9] First, the contract established that General Mills had to pay Sodima an annual licensing fee. Second, General Mills also had to pay a royalty over its gross revenues of Yoplait products in the United States.[10] The values associated with these terms are redacted in the public records and, as is generally the case with this type of contract, are unknown to researchers. However, public records do allow us to learn the structure of the licensing agreement.

In addition to these financial terms, the licensing agreement required Sodima to assist General Mills in selecting and installing equipment and machinery, planning and adjusting the production process, conducting quality control, and developing products. Under the

[9] See Yoplait Manufacturing and Distribution License Agreement. Accessed on January 29th, 2024.

[10] Though the 1977 licensing agreement was amended several times, it retained its basic structure with a licensing fee and a royalty rate. See Appendix C for the history of contract amendments.

agreement, General Mills was responsible for production and promoting the Yoplait brand in the United States.[11]

The 2011 acquisition of Yoplait by General Mills In March 2010, media reports made public that PAI Partners was considering selling its stake in Yoplait, following a change in PAI's management and a decision to redesign its investment strategy, which involved simultaneous "efforts to exit from many of its existing businesses".[12] In July 2010, PAI Partners initiated the process to sell its share in Yoplait S.A.S (the operating company) and Yoplait Marques S.A.S., which held the Yoplait brand worldwide. In February 2011, it was announced that PAI Partners had received nine bids, including bids from General Mills, Nestlé, Grupo Lala, Bright Food Group, Bel, Lactalis, Axa Private Equity, Bain Capital, and Lion Capital.[13]

Around the time PAI initiated the sale process, Sodiaal announced its intention to renegotiate the royalty rate or terminate the contract with General Mills within two years. General Mills argued that termination was possible only upon breach of contract, which Sodiaal had not alleged. In September 2010, following Sodiaal's announcement, General Mills filed for arbitration over the contract. This announcement was followed by a new one in March 2011, in which General Mills announced it would acquire PAI's share in Yoplait S.A.S and 1.5 percentage points of Sodiaal's, and 50% of Yoplait Marques S.A.S., acquiring a controlling interest in Yoplait, with Sodiaal maintaining its remaining participation in both companies.[14] The agreement between General Mills, PAI Partners, and Sodiaal was signed in May 2011 and completed on July 1st, 2011, after securing regulatory approval.[15] The transaction amount was 1.1 billion (USD).[16]

Henceforth, we assume that the transaction led to an elimination of the revenue-sharing agreement between General Mills and Sodiaal. While we believe this is a plausible assumption given that General Mills acquired a controlling stake in Yoplait, we acknowledge that the

[11] See footnote 9.

[12] "Nine bids for Yoplait as PAI speeds up disposals," Financial News London, February 8, 2011. Last accessed on December 15th, 2025.

[13] See "Private equity crashes to earth," The Sunday Times, September 20, 2009; "PAI gets nine bids for Yoplait stake," Reuters, February 6, 2011; "Nine bids for Yoplait as PAI speeds up disposals," Financial News London, February 8, 2011. Last accessed on December 15th, 2025. These sources indicate that PAI's exit strategy was not specific to Yoplait and was driven by PAI's earlier management change.

[14] See "General Mills nears $1.1 billion deal to buy half of Yoplait," The New York Times, March 18th 2011. Accessed on January 29th, 2024.

[15] See "General Mills Completes Yoplait Acquisition," BusinessWire, July 1st 2011. Accessed on January 29th, 2024.

[16] In Appendix D we show that the transaction led to minor changes in day-to-day stock returns, which is consistent with media articles that reported minor changes in General Mills' stock price (see, for example, Hughlett M. "General Mills buying biggest bite of Yoplait." Star Tribune (Minneapolis, MN). May 18, 2011. Accessed May 6, 2025.

transaction may not have fully eliminated the revenue-sharing agreement (or payments from General Mills to Sodiaal more broadly), as Sodiaal kept partial ownership of Yoplait.

Lastly, and as noted above, PAI Partners' "efforts to exit from many of its existing businesses," suggest that the sale of Yoplait was motivated by factors internal to PAI Partners that were not specific to Yoplait, as the firm was attempting to exit from "many" of its businesses at the same time.

3.2 Data

We use price and sales information on the U.S. Yogurt industry at the store-week-product level from 2010 to 2012 from the IRI Marketing Data Set. The original IRI dataset covers 64 distinct geographical areas, which correspond to groups of counties, including, generally, large metropolitan areas or regions. IRI omits markets in which the largest retailer has a market share of 50% or more (because this could reveal the retailer's identity), reducing coverage to 47 markets (Bronnenberg et al., 2008). Because of IRI's market coverage, the dataset is naturally representative of larger metropolitan areas in which no retailer accounts for more than 50% of the grocery market, suggesting that coverage is more representative in urban areas than in rural ones.

We complement these data with the 2010 American Community Survey (Ruggles et al., 2025). Given knowledge of the 5-digit ZIP codes of the IRI stores, we match households in the ACS by county and draw demographic information within the geographic areas identified in the IRI dataset. Below, we use these data to draw 500 random households from the distribution of income per person that corresponds to the store's county. We include these in our demand system.

We define a product as a brand–size combination (e.g., Yoplait Original 0.375 lbs). We measure quantities sold (and market shares) in terms of servings (a serving is defined as 0.375 lbs) at the product–store–week level. That is, one 1 lb or one 1.5 lb product is equivalent to 2.67 and 4 servings, respectively. Prices are measured as total revenue divided by total servings sold, and these are measured for each product–store–week combination.[17] We also observe a measure of in-store promotional activity at the product–store–week level, capturing whether the producer paid for an in-store advertisement display for a given product in a given store in a given week—henceforth, we use this as our measure of promotional activity. We define a market as a store–week combination, and define the market size as 1.5 times the maximum number of servings sold in that store across all weeks.

[17]The contract between General Mills and Sodima/Sodiaal defines royalties over gross sales. These are defined as General Mills's sales minus credits and returns, quantity discounts, and taxes. Our data contain retail quantities sold and revenues before taxes and net of returns.

In our analysis, we restrict attention to three firms—General Mills, Groupe Danone, and Chobani—comprising 75.6% of the category revenue during our sample period. This leaves us with 22 brands (30 products). Table A.1 in the Online Appendix presents the list of products in our sample and product-level summary statistics.

Figure A.1 in the Online Appendix presents product-level trends over time, averaging across stores, for three variables— average price per pound, average number of pounds sold, and in-store average number of promotional activity events per product. We plot these variables for the top five selling products during the sample period (in terms of revenue). The figure suggests that average prices were approximately flat across products prior to the transaction, a subject to which we return in the next section. The plots also suggest that Chobani was growing in volume sold during this period, whereas the sales of Yoplait products were declining initially but stabilized after the transaction. Lastly, the figure shows no noticeable change in in-store product-level promotional activity before or after the transaction.

4 Price Effects of Revenue-Sharing Agreements

How does the revenue-sharing agreement (RSA) impact prices? As a first approach to answering this question, we compare within-product price changes among products that were directly subject to the RSA (i.e., Yoplait products) and those that were not, before and after the RSA was discontinued. We follow a local identification strategy, from a temporal perspective, to quantify the impact of the termination of the RSA on prices. This analysis allows us to measure the impact of eliminating the RSA net of the competitive response of other firms.

The baseline specification is given by

$$\log(\text{price}_{jst}) = 1\{\text{RSA termination}\}_{jst}\beta + \eta_{js} + \phi_{ts} + \varepsilon_{jst}, \tag{1}$$

where price_{jst} is the price of product j at store s in week t, $1\{\text{RSA termination}\}_{jst}$ is an indicator that takes the value one for Yoplait products after the RSA was terminated, η_{js} and ϕ_{ts} are fixed effects at the product–store and store–week level, and ε_{jst} is an error term clustered at the store level. Note that the product and time fixed effects are allowed to vary at the store level (i.e., we include fixed effects at the product–store and week–store level). This gives the model greater flexibility in capturing that the popularity of products can vary across locations and time-varying local demand factors that can affect prices. We also present estimates using a more parsimonious set of fixed effects (i.e., product and week fixed effects) and clustering standard errors at a different level (i.e., Metropolitan Statistical

Table 1: The Impact of Revenue Sharing on Prices

	(1)	(2)	(3)	(4)
	Price (in logs)			
	Full sample		Excluding Q3/Q4 2011	
General Mills * Post July 2011	-0.026	-0.018	-0.038	-0.030
	(0.002)	(0.002)	(0.002)	(0.002)
Week FE	Yes	No	Yes	No
Week–Store FE	No	Yes	No	Yes
Product FE	Yes	No	Yes	No
Product–Store FE	No	Yes	No	Yes
N	3,213,191	3,199,520	2,671,343	2,659,893

Notes: Standard error clustered at the store level in parentheses. An observation is a store–week–product combination. Columns 3 and 4 exclude quarters 3 and 4 of 2011.

Area).

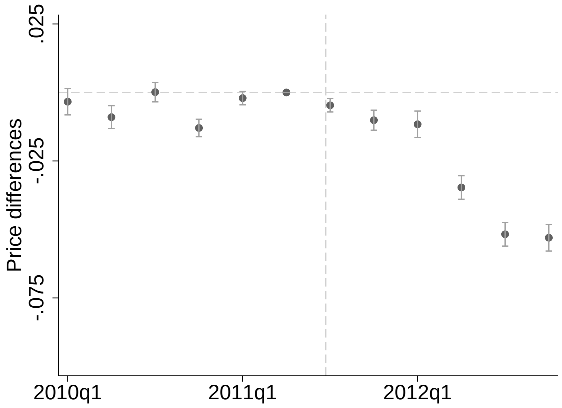
Table 1 shows estimates for Equation 1. Columns 1 and 2 make use of the full sample, whereas columns 3 and 4 exclude quarters 3 and 4 of 2011, to consider the possibility that the response to the change in incentives was delayed due to a transition period after the transaction. Column 1 shows that the prices of Yoplait products decreased by 2.6 percent relative to rival products *after* the RSA was terminated (i.e., the acquisition was completed). When using the more flexible set of fixed effects in column 2, we find that the average price effect drops to 1.8 percent. These results align with the theory suggesting that an RSA distorts incentives, leading to higher prices. Along these lines, when excluding quarters 3 and 4 of 2011 in columns 3 and 4 (to consider the possibility of a delayed change in pricing incentives), we find that the prices of Yoplait drop by 3 to 3.8 percent relative to other products after the termination of the RSA, with the smaller estimate coming from the specification with more flexible fixed effects (column 4).[18]

We also estimate a version of Equation 1 that allows for Yoplait-specific time-varying effects to examine when the price effects occurred and whether there were differential trends before the RSA was terminated. We use July 2011 (actual acquisition date) as the date for the RSA termination. Figure 1 presents the results.

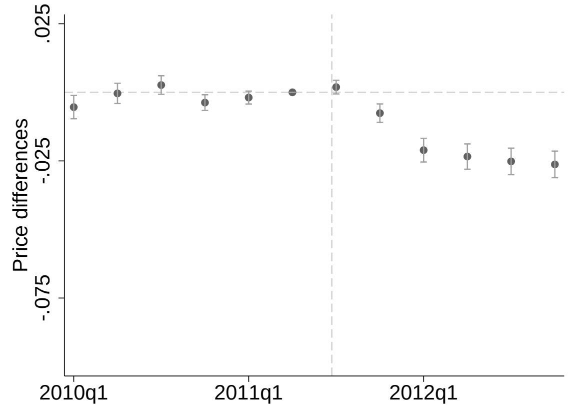
Panel A of Figure 1 considers the full sample, while Panel B restricts the sample to

[18]Table A.2 in the Online Appendix replicates Table 1 with standard errors clustered at the Metropolitan Statistical Area instead. Clustering at this level allows for prices to be correlated across products and stores within the same metropolitan area. Although standard errors increase in magnitude, the estimates remain statistically significant at conventional levels.

Figure 1: The Impact of Revenue Sharing on Prices (in Logs)



Panel A: Full sample



Panel B: No price promotions

Notes: Standard error clustered at the store level in parentheses. An observation is a store–week–product combination. All columns include product–store and store–week fixed effects.

product–store–week combinations flagged as *not* having a price promotion.[19] The vertical lines identify July 2011, when the Yoplait acquisition was completed. Both panels are aligned with Table 1 in showing an immediate price decrease in Yoplait products following the RSA termination, with the estimates reaching close to 5 or 2.5 percent by the end of the sample in panels A and B, respectively. Both panels of Figure 1 also show no evidence of differential price trends before July 2011.

That we find an immediate price decrease in Yoplait products following the RSA termination and no differential price trends gives support to the idea that the RSA termination drove the price changes and not other factors. For instance, our sample period is one where Chobani's market share is increasing over time, but Chobani's growth is gradual rather than abrupt (see Figure A.2 in the Online Appendix). While Yoplait has an incentive to lower its price to counter Chobani's growth, this incentive would manifest gradually, tracking Chobani's gradual growth, rather than abruptly, soon after the transaction. In contrast, we see no systematic changes in Yoplait prices relative to other products before the RSA termination and a sudden price drop after, making the RSA termination rather than Chobani's growth a more plausible explanation for the price effects we observe.

Lastly, we replicate Table 1 using promotional activity at the product–store–week level as the outcome variable. We do this because the RSA termination may impact incentives along other dimensions such as advertisement. Specifically, we use an indicator for whether

[19]The price promotion flag is a variable included in the IRI dataset and takes the value one when a temporary price discount of five percent or greater is in effect for a given product–store–week combination.

the producer paid for an in-store advertisement display for a given product in a given store on a given week, which is the measure of promotional activity in our dataset. We report the results in Table A.3 in the Online Appendix and find no significant effect of the RSA termination on promotional activity for Yoplait products relative to other products. This result may be an artifact of measurement error or may reflect that firms do not compete aggressively in advertising in this industry.

5 Model

5.1 Demand

We model consumers' preferences following Berry et al. (1995), Nevo (2001), and the literature that followed. We consider a market to be a store–week combination, and we model consumers' preferences in the product characteristics space and specify the indirect utility function as

$$u_{ijst} = -\alpha_i p_{jst} + \beta_i + \gamma_{js} + \gamma_t + \xi_{jst} + \varepsilon_{ijst}, \tag{2}$$

where p_{jst} and ξ_{jst} are the price and unobserved characteristics of product j at store s in week t, respectively, and γ_{js} and γ_t are product–store and time fixed effects. The consumer-specific parameter α_i is consumer i's price coefficient, whereas β_i is consumer i's taste for the inside goods (i.e., a random coefficient on the constant). Finally, we assume that ε_{ijst} is an i.i.d. extreme value type 1 idiosyncratic taste shock.

We model α_i and β_i as

$$\alpha_i = \exp\{\alpha + \sigma_p \cdot v_i^\alpha + \pi_p \cdot \text{income}_i\} \quad \text{and} \quad \beta_i = \beta + \sigma_0 \cdot v_i^\beta + \pi_0 \cdot \text{income}_i,$$

where income_i is consumer i's income (normalized to be in the unit interval) and both v_i^α and v_i^β are drawn from a standard normal distribution.[20]

5.2 Supply

To model the supply side of the market, we assume that firms compete à la Bertrand-Nash and choose the prices of their portfolio of products to maximize their profits at the market level. Specifically, we assume that firm f, selling product portfolio $\mathcal{J}_{fst}$ at store s in

[20] We specify the random coefficient on price to be log-normally distributed to lessen ex-ante restrictions on the model's cost pass-through (Miravete et al., 2023). In the case of the draws v_i^α and v_i^β, we use 500 scrambled Halton draws per market.

week t, solves

$$\max_{p_{jst} \forall j \in \mathcal{J}_{fst}} \sum_{j \in \mathcal{J}_{fst}} (p_{jst}\mu_{f(j)t} - c_{jst}) M_{st} s_{jst}(\mathbf{p}), \tag{3}$$

where $\mu_{f(j)t}$ corresponds to the share of revenues that firm f keeps (and $1-\mu_{f(j)t}$ corresponds to the share of revenues that f pays to the upstream firm), c_{jst} is the marginal cost of product j at store s in week t, and M_{st} represents the size of the market.

The first-order conditions for firm f are given by

$$s_{jst}(\mathbf{p}) + \sum_{k \in \mathcal{J}_{fst}} (p_{kst} - c_{kst}/\mu_{f(k)t}) \frac{\partial s_{kst}(\mathbf{p})}{\partial p_{jst}} = 0,$$

where we use that $\mu_{f(k)t} = \mu_{f(j)t}$ for all $j, k \in \mathcal{J}_{fst}$. We can also write the first-order conditions of market t in matrix form

$$\mathbf{s}_{st}(\mathbf{p}) - \Omega(\mathbf{p})(\mathbf{p}_{st} - \mathbf{c}_{st}/\mu_{st}) = 0, \tag{4}$$

where $\Omega(p)$ is a $J_t \times J_t$ matrix with the element-by-element product of the ownership matrix and the partial derivatives of demand with respect to prices (Nevo, 2001) and $\mathbf{c}_{st}/\mu_{st}$ is the element-wise division of $\mathbf{c}_{st}$ and μ_{st}.

Discussion In our supply-side model, we simplify the analysis by assuming that the manufacturer sets retail prices and sells directly to the end consumer. We discuss in Online Appendix B how the analysis can be extended by adding a retailer to the supply chain. As we note there, one can recover the ratios $\mathbf{c}_{st}/\mu_{st}$ (i.e., the ratios of the manufacturers' marginal costs and revenue share parameters) similarly as in Equation 4—the difference being that the equation will have extra terms to account for the retailer markup and retail price passthroughs. In our view, adding a retail sector does not enrich the economics of the analysis, which is why we choose to conduct our analysis with the parsimonious model. Other research has presented evidence suggesting that retailers play a passive role, which reassures us in our choice (see, for example, Villas-Boas, 2007; Bonnet and Dubois, 2010; Miller and Weinberg, 2017; De Loecker and Scott, 2024; Duarte et al., 2024a).

We also abstract away from the impact of the revenue-sharing agreement on firms' promotional and investment incentives. In our analysis, we find no change in Yoplait's in-store promotional activity around the termination of the agreement (see Table A.3 in the Online Appendix). That said, we acknowledge data limitations in this area. The IRI data record in-store, product–store–week promotional activity (feature and display events at the SKU level), but they do not measure other forms of advertising, nor investment in brand intangibles, which we do not observe. Our decision to abstract away from these margins is driven by the data we observe.

6 Estimation

6.1 Demand

We estimate demand using `pyBLP` (Conlon and Gortmaker, 2020). In the estimation, we follow Nevo (2001) and include product–store fixed effects ξ_{js}, which absorb time-invariant product characteristics and time-invariant product-specific demand shifters at the store level. Further, we also include week fixed effects ξ_t. Our specification includes random coefficients on price and the intercept, which allows us to rationalize substitution to the outside option more flexibly.

We estimate our demand model using a 2-step GMM approach. In the first step, we use cost shifters and local differentiation instruments to address price endogeneity and to identify the random coefficients. In the first set of instruments, we include the interaction between diesel prices and the distance from production facilities to stores (Hristakeva, 2022), an indicator that equals one for General Mills' products after the transaction between General Mills and Sodiaal and zero before it, and the mean weekly temperature in the ZIP code of each store. We also include store–week counts of products featured and displayed. We construct the second set of instruments following Gandhi and Houde (2023) and include one based on the number of products with a Euclidean distance on price that is less than one standard deviation of the price distribution, and one that interacts a product's price with this distance, which helps in identifying the random coefficient on the intercept. Finally, in the second step of our GMM estimator, we construct approximated optimal instruments following Conlon and Gortmaker (2020).

We report our estimates in Table 2. In the first column, we report estimates for the model without customer heterogeneity (i.e., $\alpha_i = \alpha$ and $\beta_i = \beta$ for all consumers i). We report the estimates of the model with customer heterogeneity in columns 2 to 4. Column 4 incorporates consumer demographics (namely, household income) into the random coefficients, whereas column 2 or 3 either do not or do so in a restricted way. The estimates in column 4 show that higher-income households are less price-sensitive and more willing to substitute toward the outside option (both effects are statistically significant). In the rest of the paper, we report results based on the estimates in column 4.

Figure A.3 in the Online Appendix reports the cumulative distribution function (CDF) of the implied own-price elasticities across all the observations in our data, using the estimates

Table 2: Demand Estimates

	(1)	(2)	(3)	(4)
Price RC (Log Normal)				
Constant (α)	1.570	1.514	1.676	2.188
	(0.011)	(0.002)	(0.005)	(0.015)
Normal draw (σ_p)		1.7E-05	5.98E-06	3.6E-04
		(0.018)	(0.010)	(0.013)
Household income (π_p)			-1.713	-9.579
			(0.043)	(0.248)
RC on Constant (Normal)				
Normal draw (σ_0)		0.036	0.021	0.050
		(0.084)	(0.040)	(0.059)
Household income (π_0)				-27.970
				(0.923)
Observations	3,199,511	3,199,511	3,199,511	3,199,511
Median own-price elasticity	-3.64	-3.44	-3.52	-3.63

Notes: Standard errors clustered at the store level in parentheses. An observation is a store–week–product combination. All specifications include product-store and week fixed effects.

in Table 2 (column 4). Table A.4 in the Online Appendix reports various percentiles of the distribution of own-price elasticities by the firm for the same specification. Overall, we find that own-price elasticities vary between -2 and -6. The median own-price elasticity is -3.63, while the mean is -3.66. These elasticities are in line with those reported in previous work but are smaller in absolute value (e.g., Villas-Boas, 2007 and Hristakeva, 2022 report mean own-price elasticity equal to -5.64 and -4.05, respectively, while Duarte et al., 2024b reports a mean own-price elasticity of -6.07 and a median of -5.58), which may be explained by our decision to estimate demand with products of the three most popular brands in this product category.[21]

[21] In Table A.5 and Table A.6 in the Online Appendix, we report the full matrices of median own- and cross-price elasticities for all the products in our data. Figure A.4 reports the distribution of the own-price elasticity for a specification that replicates that of column (4) in Table 2 but allowing the price coefficient to have a level-shift after the transaction. In this case, the overall median elasticity is -3.69; it is -3.61 before the transaction and -3.79 after, suggesting that own-price elasticities vary little over this period. Finally, to examine the stability of our estimates, we re-estimate demand on samples that extend the window back to 2009, and to 2008. Because the product–store–week demand shocks are allowed to vary flexibly while the remaining parameters are held constant within a sample, comparing estimates across windows is informative about whether preferences shifted over the period. The key parameters and the implied elasticity distribution

6.2 Revenue Sharing Agreement and Marginal Costs

Given the demand estimates, we can recover marginal costs up to the revenue-sharing agreement parameter μ using

$$\frac{\mathbf{c}_{st}}{\mu_{st}} = \mathbf{p}_{st} - \Omega(\mathbf{p}_{st})^{-1}\mathbf{s}_{st}(\mathbf{p}_{st}), \tag{5}$$

where $\mu_{jst} = \mu$ for Yoplait products prior to July 2011 and $\mu_{jst} = 1$ otherwise.[22]

The empirical challenge is separating $\mathbf{c}_{st}$ from μ_{st}. We propose two complementary approaches for doing so.

Marginal Cost Approach

Using Equation 5, we recover $\tilde{c}_{jst} \equiv c_{jst}/\mu_{jst}$ using our demand estimates. We model c_{jst} and separate it from μ_{jst} by leveraging the termination of the revenue-sharing agreement, which was specific to Yoplait products.

Specifically, we assume $c_{jst} = \exp\{\gamma_t + \kappa_j + \varepsilon_{jst}\}$, where γ_t and κ_j are week and product fixed effects, respectively, and ε_{jst} is an error term clustered at the store level. Taking the logarithm of $\tilde{c}_{jst} \equiv c_{jst}/\mu_{jst}$, and using that $\mu_{jst} = \mu$ for Yoplait products prior to July 2011 and $\mu_{jst} = 1$ otherwise, we obtain

$$\log \tilde{c}_{jst} = \gamma_t + \gamma_j - \log \mu \cdot 1\{\text{Yoplait, pre-July 2011}\} + \varepsilon_{jst}. \tag{6}$$

We assume that there is no systematic change in marginal costs except for a common trend.[23] That is, the termination of the revenue-sharing agreement is the only systematic change in $\tilde{c}_{jst} \equiv c_{jst}/\mu_{jst}$ affecting Yoplait products.

We estimate this equation using the full sample and excluding quarters 3 and 4 of 2011. Table 3 reports the estimates for the revenue sharing coefficient. We find that, when we consider the entire sample, the implied revenue-sharing coefficient is about 0.98, whereas when we exclude quarters 3 and 4 of 2011 to account for a possible transition period after the

are stable: the median own-price elasticity ranges from -3.63 (2010-2012) to -3.67 (2008-2012). Figure A.5 and Figure A.6 show that own-price elasticities are stable even over longer time periods.

[22] As mentioned in Section 3.1, we acknowledge that it is possible that the revenue-sharing agreement did not become $\mu = 1$ for Yoplait products after July 2011, but rather some figure $\mu_{\text{post-July 2011}} \geq \mu_{\text{pre-July 2011}}$. In this case, our estimate of μ captures the change in the fraction of revenue kept by General Mills prior to July 2011. Under the normalization that $\mu_{\text{post-July 2011}} = 1$, we can recover the value of $\mu_{\text{pre-July 2011}}$.

[23] Unfortunately, we do not have marginal cost data that would let us test directly for differential changes in marginal costs around the time of the transaction. However, we searched public records and media articles for changes to Yoplait's production footprint around 2011, including plant openings, closures, expansions, or downsizing, which could have impacted marginal costs, and found no evidence of such changes. Thus, we treat the absence of a systematic cost break as a maintained assumption.

Table 3: Estimating the Revenue Sharing Parameter: Marginal Cost Approach

	(1) Full sample	(2) Excluding Q3/Q4 2011
Yoplait * Pre July 2011	0.016	0.0277
	[0.0157, 0.0164]	[0.0272, 0.0282]
N	3,198,319	2,658,777
Implied Rev. Sharing Coef.	0.9841	0.9727
	[0.9837, 0.9845]	[0.9722, 0.9732]

Notes: Bootstrapped 95-percent confidence intervals in brackets. An observation is a product–store–week combination. The estimates are based on the demand system in Table 2 (column 4). In all specifications we reject the hypothesis that $\hat{\mu}$ is equal to 1.

transaction, the coefficient is about 0.97. This suggests that General Mills was paying Sodima about 2-3 percent of its revenues while the revenue-sharing agreement was in place.[24] The table reports bootstrapped 95-percent confidence intervals to account for demand parameter uncertainty.[25]

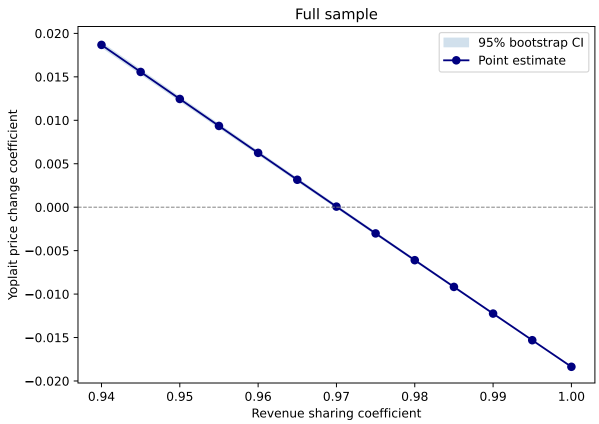
Indirect Inference Approach

In our second approach, we start by computing $\tilde{c}_{jst} \equiv c_{jst}/\mu_{jst}$ using our demand estimates. As before, we assume $\mu_{jst} = \mu$ for Yoplait products prior to July 2011 and $\mu_{jst} = 1$ otherwise. We then take an arbitrary value of μ (call it μ^{guess}) and compute the implied marginal costs: $c_{jst}^{\mu^{\text{guess}}} = \mu^{\text{guess}}\tilde{c}_{jst}$ for Yoplait products prior to July 2011 and $c_{jst}^{\mu^{\text{guess}}} = \tilde{c}_{jst}$ otherwise.

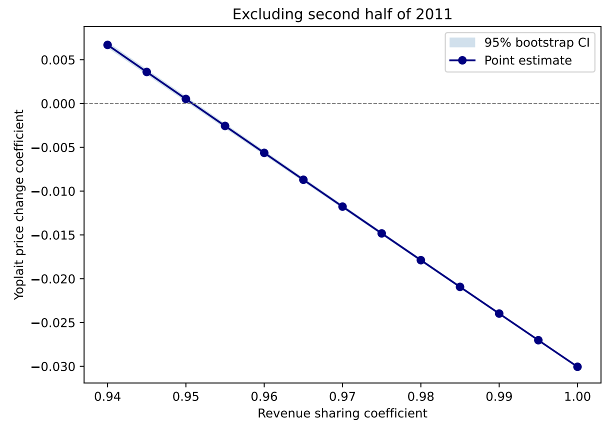
[24] In both specifications, we reject the hypothesis that $\hat{\mu}$ is equal to 1.

[25] We construct these confidence intervals using a parametric bootstrap approach. Specifically, we take 200 draws from the distribution of demand parameter estimates (based on the asymptotic approximation) and recompute marginal cost estimates given each draw. For each bootstrapped vector of marginal costs, we re-estimate Equation 6. We then form a vector of estimates for Equation 6, one for each draw from the distribution of demand parameters, and construct the confidence intervals based on this vector.

Figure 2: Estimating the Revenue Sharing Parameter: Indirect Inference Approach



A) Full sample



B) Excluding Q3/Q4 2011

Notes: The specifications make use of the estimates in Table 2 (column 4). Panel B excludes the third and fourth quarter of 2011, as in columns (3) and (4) of Table 1. The figure displays bootstrapped 95-percent confidence intervals, but these are too small to be distinguishable. See Table A.8 in the Online Appendix for details on the point estimates and confidence intervals.

Next, we compute the equilibrium prices for every market using $\mathbf{c}^{\mu^{\text{guess}}}$ and our demand estimates, assuming no revenue-sharing agreement is in place. We denote these prices $\mathbf{p}^{\mu^{\text{guess}}}$. Lastly, we estimate equation (1) using $\mathbf{p}^{\mu^{\text{guess}}}$. That is, we make within-product price comparisons of Yoplait products and products of other firms before and after July 2011, using prices $\mathbf{p}^{\mu^{\text{guess}}}$.

What is the logic of the exercise? The equilibrium prices $\mathbf{p}^{\mu^{\text{guess}}}$ are computed under the assumption that no revenue-sharing agreement is in place. Thus, the cost adjustment given by $c_{jst}^{\mu^{\text{guess}}} = \mu^{\text{guess}}\tilde{c}_{jst}$ evaluated at the true value of μ should strip equilibrium prices of any distortion induced by the revenue-sharing agreement, assuming the revenue-sharing agreement termination was the only systematic factor affecting the industry through July 2011. In other words, the estimates of equation (1) using $\mathbf{p}^{\mu^{\text{guess}}}$ should reveal a null effect (i.e., $\hat{\beta} = 0$ in equation 1) if the value μ^{guess} equals the true value of μ. Using this logic, we repeat the steps above, searching for the value of μ^{guess} such that $\hat{\beta} = 0$.

Figure 2 presents the estimates of β in equation (1) using the prices $\mathbf{p}^{\mu^{\text{guess}}}$ that correspond to different values of μ^{guess}, along with bootstrapped confidence intervals.[26] We do this

[26] We construct these confidence intervals using a parametric bootstrap approach. Specifically, we take 200 draws from the distribution of demand parameter estimates (based on the asymptotic approximation) and recompute marginal cost estimates given each draw. We implement the procedure described above for each bootstrapped vector of marginal costs, one for each draw from the distribution of demand parameters, and compute the confidence intervals based on the bootstrapped distribution of estimates. See Table A.8 for details.

exercise using the demand estimates in Table 2 (column 4), and we repeat the analysis excluding quarters 3 and 4 of 2011. In Figure 2.A we find that the value of μ that leads to $\hat{\beta} = 0$ is $\hat{\mu} = 0.97$; whereas in Figure 2.B, when excluding quarters 3 and 4 of 2011, we find that this value is $\hat{\mu} = 0.95$. These results suggest that General Mills paid Sodima between 3 and 5 percent of its revenues while the revenue-sharing agreement was in place.

Discussion

We conclude this section by noting that both methods rely on the assumption that the termination of the RSA was the one systematic factor that changed pricing incentives during the sample period. Along these lines, the marginal cost approach imposes an explicit constraint on the evolution of marginal costs through the termination of the RSA—i.e., marginal costs are only allowed to change over time through a common trend.

The key difference among both methods is that the marginal cost approach uses a transformation of the equilibrium vector of prices (see equation 5) whereas the indirect inference approach uses prices directly. Confounders that impact prices (e.g., a product-specific demand shock) may have a differential effect on the estimates of the RSA depending on the method used.

We find similar estimates across methods ($\hat{\mu} = 0.95$ and $\hat{\mu} = 0.97$), which suggests that confounders that violate our identification assumption (if any) play a minor role. That said, the marginal cost approach delivers slightly larger estimates than the indirect inference approach. Why?

As noted above, the sample period is one where Chobani's market share is growing and its prices are flat or falling (see Figure A.1 in the Online Appendix), which suggests in part that Chobani's marginal costs are falling, making the marginal cost approach's assumption of no differential cost changes over time a poor fit for Chobani during the sample period. For this reason, we replicate Table 3 dropping Chobani from the estimation sample and present the results in Table A.7 in the Online Appendix. The table shows that when we drop Chobani, the marginal cost approach delivers estimates of the revenue sharing coefficient nearly identical to the estimates from the indirect inference approach in Figure 2. These results highlight the extra rigidity of the marginal cost approach.

7 The Equilibrium Effects of Revenue-Sharing Agreements

How does the revenue-sharing agreement (RSA) impact General Mills' profits and supply chain? We answer this question by comparing the baseline equilibrium (i.e., that with the RSA in place) with the counterfactual equilibrium in which Yoplait's supply chain is vertically integrated (i.e., the revenue-sharing agreement parameter is set to $\mu = 1$). This comparison allows us to learn about the equilibrium impacts of the RSA, considering both the distortion on pricing incentives as well as the potential easing of competition intensity.

Throughout this section, we abstract from the fact that Sodiaal retained a partial ownership stake in Yoplait after General Mills' acquisition of a controlling interest in Yoplait in July 2011, and that some payments may have flowed from General Mills to Sodiaal thereafter. We also restrict attention to the period before July 2011.

General Mills's Outcomes

We first measure the impact of the RSA on endogenous outcomes set by General Mills. We compute and compare these endogenous variables under three scenarios: i) baseline equilibrium with RSA; ii) no RSA, but General Mills is the only firm allowed to reoptimize; and iii) equilibrium without RSA. The difference between scenarios ii) and iii) allows us to separate the direct effect of eliminating the RSA from the equilibrium feedback effects that come into play once all firms best respond and a new equilibrium without RSA is reached.

Table 4 presents the results of this comparison for the model estimates based on Table 2 (column 4).[27] We consider RSAs with values of $\hat{\mu} = 0.97$ (estimate using the marginal cost approach) and $\hat{\mu} = 0.95$ (estimate using the indirect inference approach). Each column of the table presents estimates for the following equation

$$\log(X_{jswt}) = \delta_0 + 1\{\text{no RSA, best-response}\}_{jswt}\delta_1 + 1\{\text{no RSA, equilibrium}\}_{jswt}\delta_2 + \varepsilon_{jswt},$$

where X_{jswt} is an outcome variable (price, market share, or profit) for product j at store s in week w under scenario $t \in \{$baseline equilibrium; no RSA–best-response; no RSA–equilibrium$\}$. The omitted category in the equation is the baseline equilibrium scenario.

The table shows that removing the revenue-sharing agreement leads to an average equilibrium price decrease of 1.9 to 3.1 percent for Yoplait (General Mills) products, depending on the value of μ. As discussed in Section 2, from the perspective of pricing incentives, an

[27]See Table A.9 in the Online Appendix for more details, with a breakdown of changes in market outcomes by firm.

Table 4: The Impact of a Revenue Sharing Agreement on Market Outcomes: General Mills

	(1)	(2)	(3)	(4)	(5)	(6)
		$\mu = 0.97$			$\mu = 0.95$	
	Price	Share	Profit	Price	Share	Profit
Best-response	-0.0188	0.0619	0.0941	-0.0314	0.1029	0.1574
wo/RSA	(0.0005)	(0.0022)	(0.0021)	(0.0004)	(0.0021)	(0.0021)
Equilibrium	-0.0188	0.0618	0.0940	-0.0314	0.1028	0.1572
wo/RSA	(0.0005)	(0.0022)	(0.0021)	(0.0004)	(0.0021)	(0.0021)
N	2,402,718	2,402,718	2,402,718	2,402,718	2,402,718	2,402,718

Notes: The estimates are based on demand estimates in Table 2 (column 4). Standard errors in parentheses. An observation is a product–store–week combination. We restrict attention to weeks before July 2011 (i.e., the time when the transaction was completed) and General Mills products. Each column displays regression coefficients of $\log(X)$, for $X \in \{\text{price}, \text{market share}, \text{profit}\}$, on an indicator for when General Mills best responds to the removal of the RSA (but all other firms are not allowed to respond) and an indicator for the equilibrium without RSA (an indicator for the baseline equilibrium is the omitted category).

RSA is equivalent to having an inflated marginal cost. When the RSA is eliminated, the upward pressure on prices introduced by the RSA is eliminated.

Did downstream competition limit the extent to which the RSA introduced inefficiencies in this market? The model we discussed in Section 2 shows that the main effect of the RSA is to distort the pricing incentives of the firm that is subject to it. However, the presence of downstream rivals that are not subject to the RSA limits the extent of the inefficiency caused by this contract, as these firms may respond by raising their prices as well. Table 4 shows that the competitive effects are of second order in this setting—the effects on prices are nearly identical between the cases in which the RSA is eliminated and only General Mills is allowed to reoptimize (row labeled 'Best-response wo/RSA') and the equilibrium without the RSA (row labeled 'Equilibrium wo/RSA').

We find that eliminating the RSA causes an average increase in the equilibrium market share of General Mills products of between 6.2 to 10.3 percent. This suggests that eliminating the RSA helps General Mills increase the market share of its products by stealing market share away from rivals and the outside option—a result consistent with Figure A.1 in the Online Appendix.

Lastly, we compute the change in a product's profit in a store when eliminating the RSA. We find that the equilibrium profits of Yoplait products, captured by General Mills, increased by 9.4 to 15.7 percent on average. These numbers combine two effects. On the one hand, the termination of the RSA implies that General Mills no longer has to pay Sodiaal part of its revenue, which induces a mechanical profit gain.[28] On the other hand, the termination of the RSA implies that General Mills becomes more competitive in the product market game (i.e., it can compete as though it has "lower" marginal costs). Being more competitive increases equilibrium profits. We find that both effects are empirically relevant, but the former accounts for about 95 to 96 percent of the gains reported in Table 4.[29]

How large are these effects? We find that eliminating the RSA would have increased the profit earned by General Mills in a store–year by between \$957 ($\mu = 0.97$) and \$1,612 ($\mu = 0.95$) on average. Assuming that the sample of stores in the IRI Marketing Data Set is representative of the universe of supermarket/grocery stores in the U.S. at the time of General Mills's acquisition of Yoplait, the nation-wide change in General Mills' annual profits from eliminating the RSA is between \$62 million and \$104 million.[30] Note that we abstract away from licensing fees from this calculation (i.e., lump sum payments between General Mills and Sodima).

Two facts are consistent with the transaction adding value. First, given the \$1.1 billion transaction price, the profit gains estimated above imply a rate of return on the order of 5.6 to 9.5 percent, comparable to the reciprocal of General Mills' price-earnings ratio at the time (in the 14–15 range; see Online Appendix D). Second, General Mills' stock price rose by 1.44 percent on the announcement,[31] which is consistent with the deal adding value to General Mills, though, as we show in Online Appendix D, it is not statistically distinguishable from the firm's normal daily returns.

General Mills's Supply Chain Outcomes

Our findings suggest a significant distortion in prices and the profits of General Mills caused by the revenue-sharing agreement (RSA). What is the impact of the RSA on Yoplait's supply chain profits? In the scenario with the RSA, the profits of the supply chain equal the sum of the profits of General Mills (product market profits minus the revenue shared with Sodiaal) and Sodiaal (revenue collected from General Mills), whereas in the scenario without the RSA the profits of the supply chain equal the profits of General Mills.

As argued in Section 2, the RSA has two effects on the supply chain's profits. On the one

[28]As mentioned at the beginning of the section, we abstract away from any payments flowing from General Mills to Sodiaal that may exist due to Sodiaal's remaining stake in Yoplait.

[29]Specifically, 96 percent ($\mu = 0.97$) and 95 percent ($\mu = 0.95$) of the profit gains in Table 4 result from terminating the payments from General Mills to Sodiaal. The rest of the profit gains come from more competitive pricing.

[30]Here, we scale up the store–year average profit change (between \$957 and \$1,612) by 64,366, which is the number of supermarket/grocery stores in the U.S. in 2011 according to the County Business Patterns.

[31]Mike Hughlett, "General Mills confirms deal to buy 50% of Yoplait for \$1.1B", *StarTribune*, March 19th, 2011. Accessed through the Texas A&M University Libraries on August 20th, 2025.

Table 5: The Impact of a Revenue Sharing Agreement on General Mills' Supply Chain Profit

	(1) $\mu = 0.97$	(2) $\mu = 0.95$
	General Mills's Supply Chain Profit	
Best-response wo/RSA	0.0039 (0.0021)	0.0086 (0.0021)
Equilibrium wo/RSA	0.0038 (0.0021)	0.0085 (0.0021)
N	2,402,718	2,402,718

Notes: The estimates are based on demand estimates in Table 2 (column 4). Standard errors in parentheses. An observation is a product–store–week combination. We restrict attention to weeks before July 2011 (i.e., the time when the transaction was completed) and General Mills products. Each column displays regression coefficients of log(Profit of supply chain) on an indicator for when General Mills best responds to the removal of the RSA (but all other firms are not allowed to respond) and an indicator for the equilibrium without RSA (an indicator for the baseline equilibrium is the omitted category).

hand, it inflates prices, which increases the price–cost margin of the supply chain. On the other hand, the greater prices cause a decrease in the market shares of the supply chain's products. Depending of the relative magnitude of these effects, the RSA may increase or decrease supply chain profits.

Table 5 presents the results of comparing supply chain profits across the same scenarios as in Table 4. The table shows that eliminating the RSA increases General Mills' supply chain profits by 0.38 and 0.85 percent when $\mu = 0.97$ and $\mu = 0.95$, respectively. These findings suggest that while the RSA has an economically significant impact on General Mills' profits, it has a modest impact on the supply chain's profits.

As discussed in Section 2, the elasticity of demand plays a role in explaining why the RSA does not lead to a significant easing of the intensity of price competition, which is necessary for the RSA to increase joint supply chain profits. Given the estimated demand elasticities and broader market conditions, the RSA's inefficiency effect dominates, explaining the result that eliminating the RSA yields a net gain in joint supply chain profits.

8 Concluding Remarks

Revenue-sharing agreements (RSAs) are common in vertical contracting, yet their equilibrium effects on joint supply chain profits are theoretically ambiguous. Empirical evidence informing the question of their equilibrium impacts is limited due to the confidentiality of business-to-business contract terms.

In this paper, we develop a structural framework to identify the terms of a revenue-sharing agreement, exploiting variation in vertical structure in conjunction with an equilibrium model of supply and demand for the U.S. yogurt industry. Using two complementary identification strategies, we estimate that the downstream firm (General Mills) remitted between 3 and 5 percent of its annual Yoplait revenues to the upstream firm (Sodiaal) under the RSA.

Having recovered the terms of the RSA, we proceed to quantify the equilibrium impacts of the vertical structure. We find that the transition from the RSA to full vertical integration eliminated the contract's upward pricing distortion, resulting in a retail price reduction of 1.9 to 3.1 percent for Yoplait products. Eliminating the RSA did not significantly increase price competition intensity, as the RSA only had a second-order effect on rival firms' prices.

We find that the RSA had a negative impact on the joint supply chain profits of 0.4 to 0.85 percent. These estimates suggest that the inefficiency effect outweighed the reduction in competition intensity, making the RSA a surplus-decreasing contract for the Yoplait supply chain.

We also find that eliminating the RSA increased General Mills' equilibrium profits by 9.4 to 15.7 percent, reflecting gains from efficient pricing and the elimination of revenue sharing with Sodiaal. That is, General Mills' vertical integration may be explained by an effort to eliminate pricing distortions that put it at a competitive disadvantage relative to rivals (especially Chobani, which was seeing growth), as well as by an attempt to fully appropriate the Yoplait vertical rents.

Lastly, the estimate that the RSA was surplus-decreasing for the supply chain by 0.4–0.85 percent gives a lower bound on the monitoring and incentive issues along the supply chain that motivated the use of a RSA. We believe that quantifying the economic consequences of misaligned incentives along the supply chain is a contribution in and of itself.

As well, that this loss is small (0.4 to 0.85 percent of joint supply chain profits) helps explain why the contract persisted, largely unchanged, for over three decades: the surplus at stake was likely modest relative to the cost of renegotiating, and may also reflect that no feasible alternative contract addressed all relevant incentive issues at a lower cost. Ultimately, it was vertical integration—at a transaction price of $1.1 billion—that eliminated the distortion.

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Online Appendix: Not For Publication

The Equilibrium Effects of Revenue-Sharing Agreements

Fernando Luco and Guillermo Marshall

A Additional Tables and Figures

Table A.1: Summary Statistics

Brand	Firm	0.375 lbs			1 lbs			1.5 lbs		
		N	Mean	S.D.	N	Mean	S.D.	N	Mean	S.D.
Brown Cow	Groupe Danone	35,273	1.06	0.24	9,132	3.72	0.56			
Chobani	Chobani	166,203	1.33	0.20	78,602	3.40	0.52			
Dannon	Groupe Danone	115,589	0.67	0.18						
Dannon Activia	Groupe Danone				199,994	2.48	0.42	62,085	3.00	0.48
Dannon Activia Dessert	Groupe Danone				40,468	2.33	0.44			
Dannon Activia Fiber	Groupe Danone				123,824	2.45	0.43			
Dannon Activia Light	Groupe Danone				188,986	2.47	0.42	51,413	2.90	0.51
Dannon All Natural	Groupe Danone	131,621	0.66	0.15						
Dannon Danimals Crushcups	Groupe Danone				127,910	2.36	0.34			
Dannon Lght N Ft Crb & Sugr C	Groupe Danone				67,062	3.05	0.37			
Dannon Light N Fit	Groupe Danone	132,490	0.66	0.16				177,785	2.21	0.35
Dannon Nutriday	Groupe Danone				4,976	1.01	0.08			
Stonyfield Farm	Groupe Danone	85,092	0.96	0.20				20,811	4.08	0.60
Yoplait	General Mills	138,101	1.04	0.21	42,329	2.85	0.44			
Yoplait Delights	General Mills				119,754	2.87	0.46			
Yoplait Fiber One	General Mills				87,553	2.52	0.42			
Yoplait Light	General Mills	203,545	0.67	0.13				0		
Yoplait Light Thick & Creamy	General Mills	141,848	0.66	0.12						
Yoplait Original	General Mills	212,541	0.69	0.15				96,226	2.47	0.42
Yoplait Thick & Creamy	General Mills	150,554	0.66	0.12						
Yoplait Trix	General Mills							130,972	2.69	0.46
Yoplait Yo Plus	General Mills				70,452	2.36	0.45			

Notes: An observation is a store–week–product combination. The columns labeled "Mean" and "S.D." report the mean and standard deviation of prices for each product.

Table A.2: The Impact of Revenue Sharing on Prices: Clustering Standard Errors at the MSA Level

	(1)	(2)	(3)	(4)
	Price (in logs)			
	Full sample		Excluding Q3/Q4 2011	
General Mills * Post July 2011	-0.026	-0.018	-0.038	-0.030
	(0.005)	(0.005)	(0.006)	(0.006)
Week FE	Yes	No	Yes	No
Week–Store FE	No	Yes	No	Yes
Product FE	Yes	No	Yes	No
Product–Store FE	No	Yes	No	Yes
N	3,213,191	3,199,520	2,671,343	2,659,893

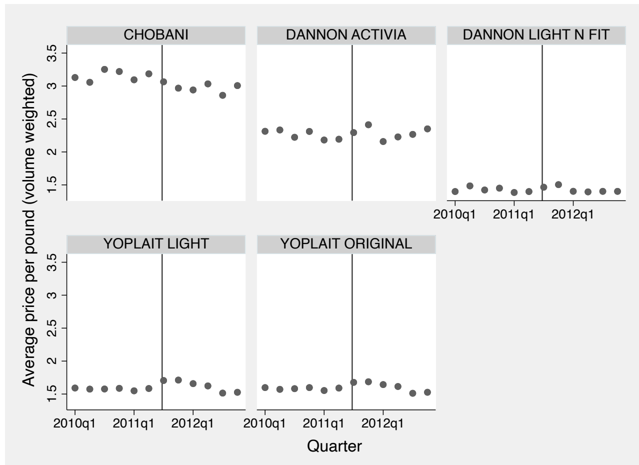
Notes: Standard error clustered at the Metropolitan Statistical Area level in parentheses. An observation is a store–week–product combination. Columns 3 and 4 exclude quarters 3 and 4 of 2011.

Table A.3: The Impact of Revenue Sharing on Promotional Activity

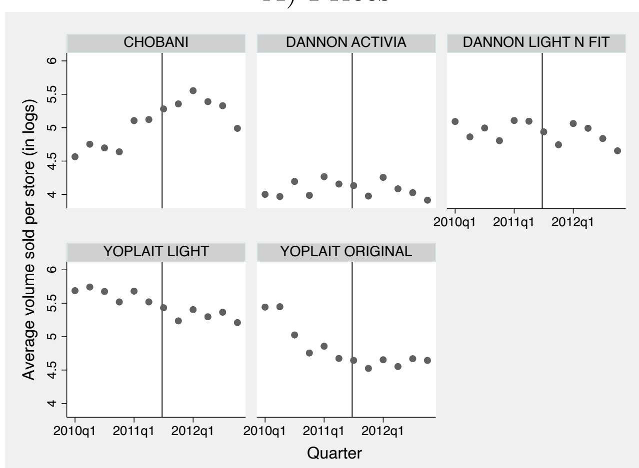
	(1)	(2)	(3)	(4)
	Indicator for minor or major advertisement/display event			
	Full sample		Excluding Q3/Q4 2011	
General Mills * Post July 2011	-.0000681	5.71e-06	-.000154	-.000068
	(.0000843)	(.0000693)	(.000118)	(.000106)
Week FE	Yes	No	Yes	No
Week–Store FE	No	Yes	No	Yes
Product FE	Yes	No	Yes	No
Product–Store FE	No	Yes	No	Yes
N	3,213,192	3,199,521	2,671,343	2,659,893

Notes: Standard error clustered at the store level in parentheses. An observation is a store–week–product combination. Minor or major advertisement/display events are flagged in the dataset at the store–week–product level. Columns 3 and 4 exclude quarters 3 and 4 of 2011.

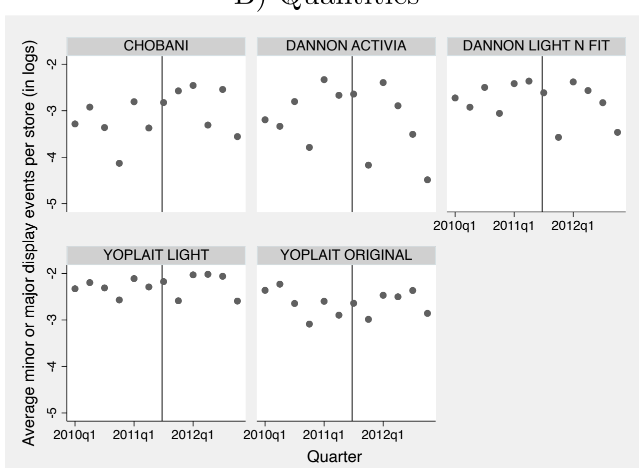
Figure A.1: Price, quantity, and volume trends for the top five products (in revenue)



A) Prices



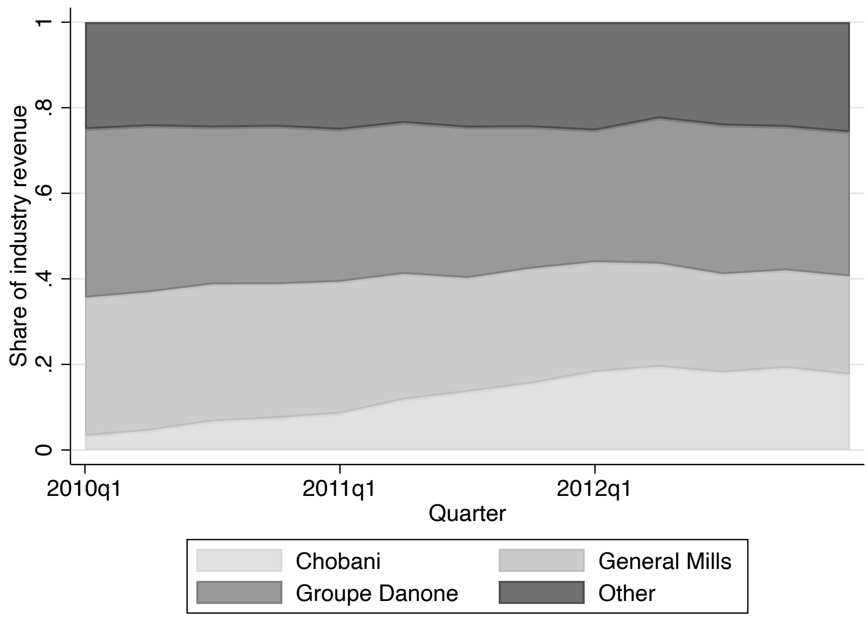
B) Quantities



C) Promotional activity

Notes: The figure reports the average volume-weighted price per pound for each product across stores, average volume (in pounds) sold for each product across stores, and average number of minor or major display events per product (e.g., end of aisle displays) across stores. We present these for the five products with the greatest cumulative revenue during the sample period. The vertical line indicates the time of the transaction, i.e., quarter 3 of 2011.

Figure A.2: Share of Industry Revenue, by Firm



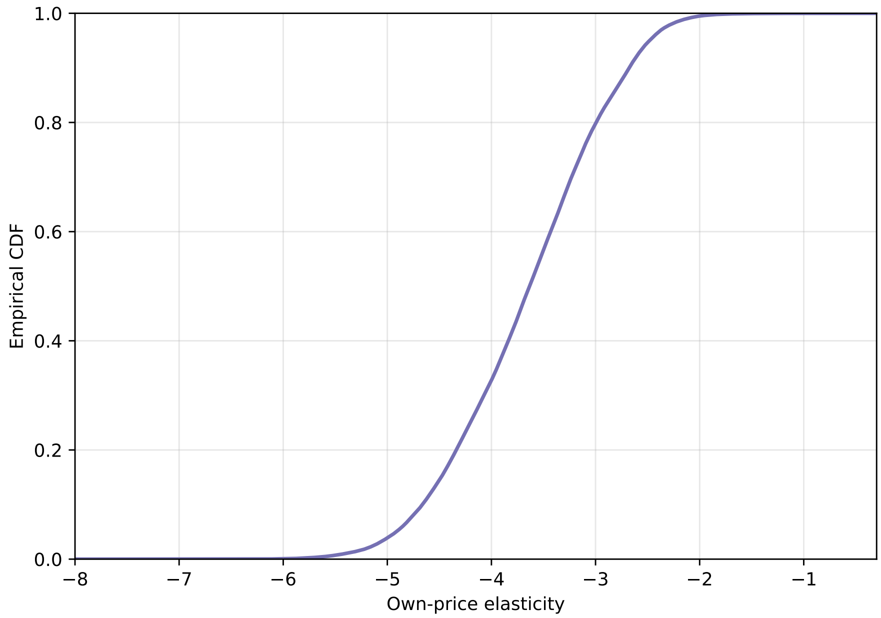
Notes: The figure reports the share of industry revenue captured by each firm in a quarter. We compute industry revenue shares using all products by all firms, including those that are otherwise excluded from our analysis (e.g., private labels).

Table A.4: Estimated own-price elasticities by firm

		Percentile								
Firm	Mean	1%	5%	10%	25%	50%	75%	90%	95%	99%
Chobani	-4.56	-5.93	-5.57	-5.36	-4.95	-4.56	-4.16	-3.80	-3.61	-3.21
General Mills	-3.52	-5.20	-4.81	-4.55	-4.00	-3.42	-3.02	-2.66	-2.48	-2.16
Danone	-3.64	-5.18	-4.77	-4.55	-4.15	-3.68	-3.16	-2.64	-2.45	-2.07

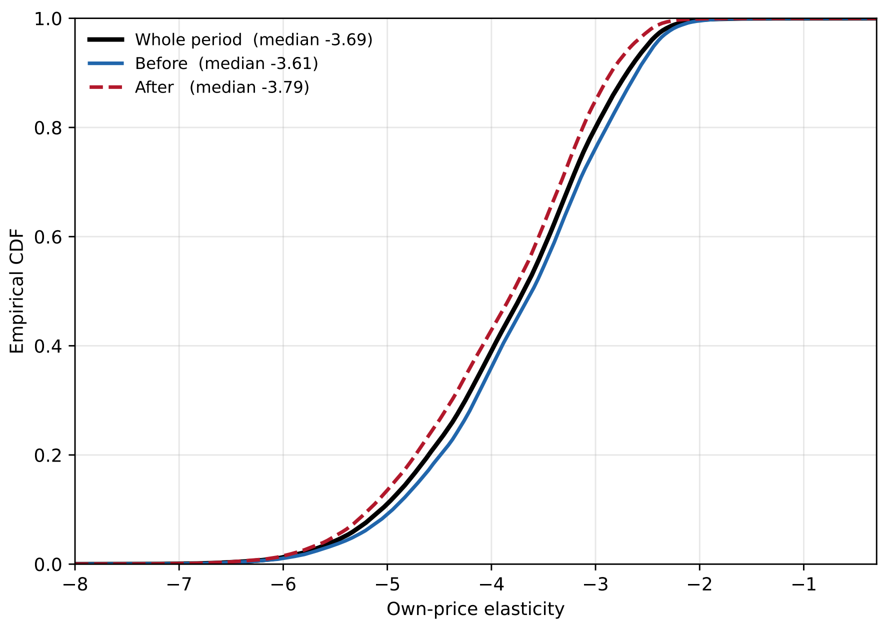
Notes: The table reports percentiles of the firm-specific distribution of own-price elasticities. An observation is a product-store-week combination. The table is based on the estimates of specification 4 in Table 2

Figure A.3: Estimated own-price elasticities



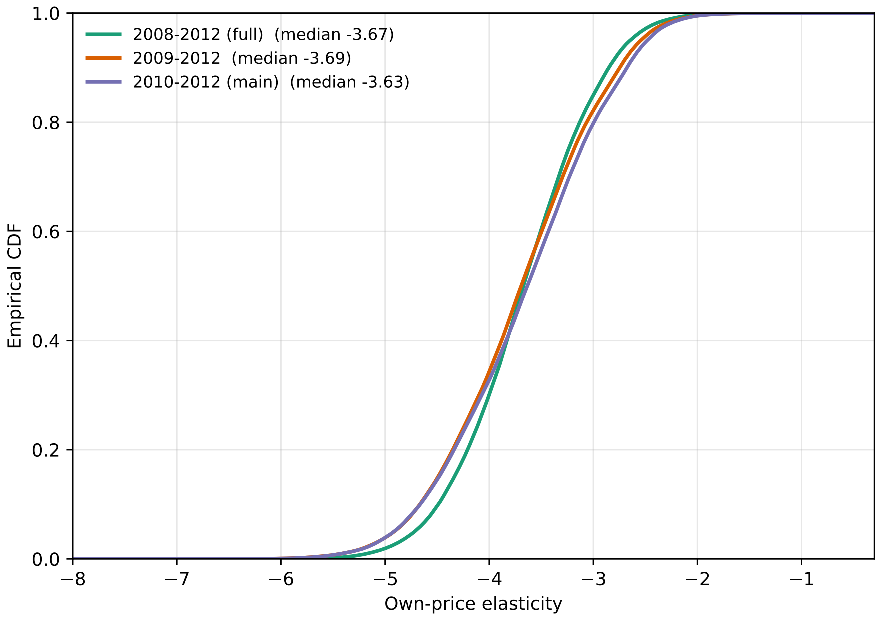
Notes: The figure reports the cumulative distribution function (CDF) of the estimated own-price elasticities according to specification 4 in Table 2.

Figure A.4: Estimated own-price elasticities allowing the price coefficient to vary over time



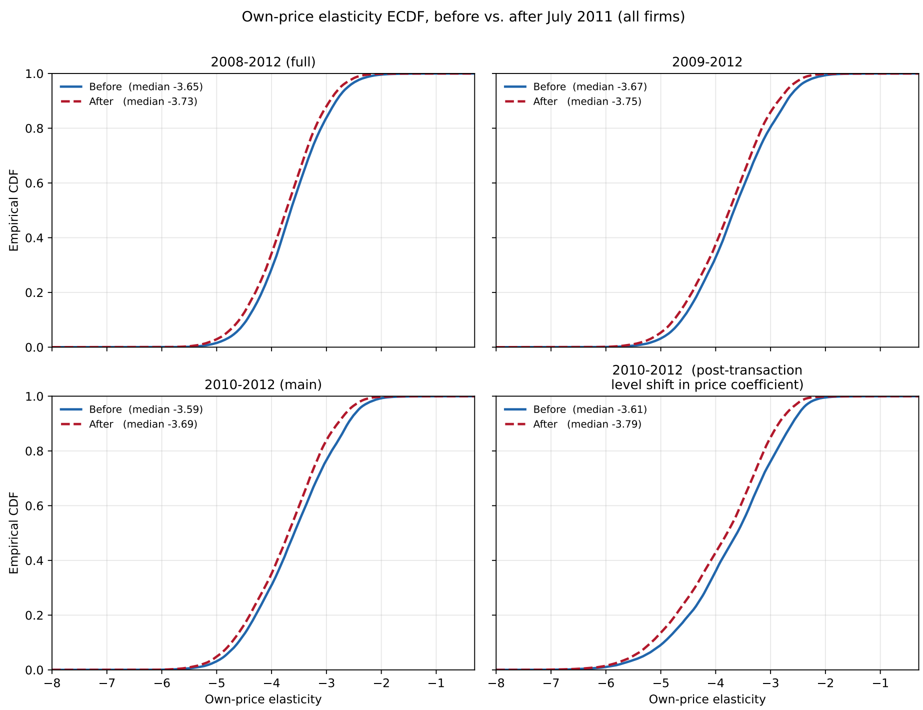
Notes: The figure reports the cumulative distribution function (CDF) of the estimated own-price elasticities for a specification that replicates specification 4 in Table 2 but that allows the price coefficient to have a level shift after the transaction.

Figure A.5: Estimated own-price elasticities for different sample periods



Notes: The figure reports the cumulative distribution function (CDF) of the estimated own-price elasticities for a specification that replicates specification 4 in Table 2 but using three sample periods.

Figure A.6: Estimated own-price elasticities, before and after the transaction, by sample period



Notes: The figure reports the cumulative distribution function (CDF) of the estimated own-price elasticities for four samples, before and after the transaction, according to specifications 4-6 in Table 2. The fourth plot is the 2010-2012 sample, but the demand specification allows the price coefficient to vary by including a dummy equal to one after the transaction.

Table A.5: Estimated median own- and cross-price elasticities (Log Normal specification)

		ID														
ID	Brand	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	Brown Cow 0.375	-4.250	0.008	0.184	0.022	0.033	0.065	0.005	0.004	0.013	0.042	0.005	0.008	0.007	0.009	0.056
2	Brown Cow 1.0	0.005	-4.682	0.186	0.016	0.024	0.042	0.004	0.003	0.009	0.027	0.004	0.008	0.009	0.007	0.069
3	Chobani 0.375	0.010	0.006	-4.545	0.020	0.024	0.059	0.004	0.004	0.012	0.033	0.004	0.007	0.007	0.008	0.044
4	Chobani 1.0	0.008	0.006	0.250	-4.583	0.024	0.053	0.004	0.003	0.011	0.030	0.003	0.009	0.007	0.008	0.050
5	Dannon 0.375	0.005	0.005	0.171	0.016	-3.186	0.058	0.006	0.004	0.012	0.033	0.005	0.011	0.008	0.007	0.063
6	Dannon Activia 1.0	0.011	0.006	0.168	0.018	0.027	-3.963	0.005	0.004	0.014	0.038	0.005	0.008	0.009	0.008	0.051
7	Dannon Activia 1.5	0.010	0.006	0.137	0.016	0.033	0.057	-3.582	0.003	0.011	0.033	0.005	0.010	0.008	0.007	0.064
8	Dannon Activia Dessert 1.0	0.007	0.005	0.130	0.017	0.031	0.074	0.005	-3.833	0.013	0.041	0.005	0.009	0.008	0.008	0.063
9	Dannon Activia Fiber 1.0	0.011	0.006	0.158	0.018	0.030	0.069	0.005	0.004	-3.959	0.040	0.005	0.009	0.009	0.008	0.058
10	Dannon Activia Light 1.0	0.011	0.006	0.168	0.018	0.027	0.067	0.005	0.004	0.014	-3.990	0.005	0.008	0.009	0.008	0.051
11	Dannon Activia Light 1.5	0.008	0.006	0.125	0.016	0.032	0.063	0.005	0.004	0.013	0.037	-3.512	0.010	0.009	0.007	0.064
12	Dannon All Natural 0.375	0.010	0.005	0.165	0.016	0.033	0.062	0.006	0.004	0.013	0.037	0.006	-3.198	0.008	0.007	0.065
13	Dannon Danimals Crushcups 1.0	0.009	0.006	0.148	0.017	0.031	0.066	0.005	0.004	0.014	0.038	0.005	0.009	-3.902	0.008	0.058
14	Dannon Light N FT CRB & Sugar 1.0	0.008	0.007	0.235	0.020	0.028	0.054	0.004	0.003	0.011	0.030	0.004	0.010	0.007	-4.430	0.056
15	Dannon Light N Fit 0.375	0.007	0.005	0.152	0.016	0.032	0.059	0.006	0.004	0.012	0.033	0.005	0.011	0.008	0.007	-3.143
16	Dannon Light N Fit 1.5	0.010	0.004	0.132	0.015	0.032	0.066	0.006	0.004	0.014	0.037	0.006	0.010	0.009	0.007	0.059
17	Dannon Nutriday 1.0	0.006	0.002	0.054	0.012	0.024	0.084	0.009	0.007	0.023	0.052	0.012	0.010	0.019	0.008	0.046
18	Stonyfield Farm 0.375	0.008	0.006	0.185	0.017	0.032	0.061	0.005	0.004	0.013	0.038	0.005	0.010	0.009	0.008	0.067
19	Stonyfield Farm 1.5	0.009	0.005	0.259	0.017	0.032	0.053	0.004	0.003	0.011	0.034	0.004	0.012	0.008	0.008	0.073
20	Yoplait 0.375	0.009	0.006	0.173	0.018	0.024	0.061	0.005	0.004	0.012	0.035	0.005	0.007	0.007	0.007	0.044
21	Yoplait 1.0	0.011	0.007	0.193	0.016	0.022	0.059	0.004	0.003	0.009	0.034	0.004	0.006	0.004	0.007	0.033
22	Yoplait Delights 1.0	0.010	0.006	0.157	0.018	0.027	0.062	0.005	0.004	0.013	0.037	0.005	0.008	0.008	0.008	0.052
23	Yoplait Fiber One 1.0	0.008	0.006	0.145	0.017	0.029	0.065	0.005	0.004	0.013	0.039	0.005	0.009	0.009	0.008	0.059
24	Yoplait Light 0.375	0.010	0.005	0.145	0.016	0.030	0.068	0.006	0.004	0.014	0.037	0.006	0.009	0.009	0.007	0.057
25	Yoplait Light Thick & Creamy 0.375	0.010	0.005	0.128	0.016	0.031	0.065	0.006	0.004	0.014	0.037	0.006	0.010	0.009	0.007	0.058
26	Yoplait Original 0.375	0.010	0.005	0.147	0.016	0.031	0.068	0.006	0.004	0.014	0.037	0.006	0.009	0.009	0.007	0.057
27	Yoplait Original 1.5	0.010	0.004	0.142	0.014	0.029	0.066	0.005	0.004	0.013	0.036	0.005	0.008	0.006	0.007	0.049
28	Yoplait Thick & Creamy 0.375	0.010	0.005	0.144	0.016	0.031	0.066	0.006	0.004	0.013	0.038	0.006	0.010	0.009	0.007	0.058
29	Yoplait Trix 1.5	0.010	0.005	0.119	0.015	0.034	0.069	0.006	0.004	0.015	0.040	0.006	0.010	0.010	0.007	0.065
30	Yoplait Yo Plus 1.0	0.009	0.006	0.115	0.018	0.029	0.064	0.006	0.004	0.014	0.039	0.006	0.009	0.011	0.008	0.061

Table A.6: Estimated median own- and cross-price elasticities (Log Normal specification)

		ID														
ID	Brand	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1	Brown Cow 0.375	0.042	0.005	0.018	0.021	0.017	0.009	0.014	0.010	0.154	0.012	0.114	0.011	0.011	0.017	0.010
2	Brown Cow 1.0	0.023	0.000	0.032	0.016	0.013	0.006	0.014	0.010	0.092	0.013	0.065	0.006	0.013	0.019	0.009
3	Chobani 0.375	0.038	0.006	0.016	0.010	0.014	0.007	0.013	0.009	0.104	0.011	0.080	0.009	0.010	0.022	0.008
4	Chobani 1.0	0.042	0.003	0.021	0.010	0.011	0.005	0.012	0.009	0.088	0.011	0.067	0.008	0.012	0.024	0.008
5	Dannon 0.375	0.056	0.013	0.018	0.010	0.012	0.006	0.013	0.010	0.127	0.018	0.093	0.012	0.015	0.032	0.010
6	Dannon Activia 1.0	0.047	0.013	0.016	0.010	0.015	0.007	0.014	0.010	0.136	0.015	0.107	0.012	0.012	0.028	0.010
7	Dannon Activia 1.5	0.042	0.008	0.018	0.010	0.014	0.007	0.012	0.010	0.131	0.019	0.097	0.010	0.014	0.029	0.010
8	Dannon Activia Dessert 1.0	0.048	0.004	0.019	0.011	0.018	0.009	0.015	0.010	0.150	0.017	0.111	0.012	0.014	0.028	0.009
9	Dannon Activia Fiber 1.0	0.045	0.012	0.018	0.010	0.016	0.007	0.015	0.011	0.135	0.017	0.108	0.012	0.014	0.029	0.011
10	Dannon Activia Light 1.0	0.047	0.012	0.016	0.010	0.015	0.007	0.014	0.010	0.137	0.015	0.105	0.012	0.012	0.027	0.010
11	Dannon Activia Light 1.5	0.042	0.013	0.018	0.008	0.016	0.007	0.014	0.011	0.155	0.019	0.110	0.012	0.014	0.028	0.012
12	Dannon All Natural 0.375	0.054	0.013	0.018	0.010	0.013	0.007	0.012	0.010	0.145	0.017	0.107	0.012	0.014	0.029	0.010
13	Dannon Danimals Crushcups 1.0	0.046	0.013	0.019	0.010	0.017	0.007	0.014	0.011	0.138	0.017	0.108	0.011	0.014	0.029	0.011
14	Dannon Light N FT CRB & Sugar 1.0	0.042	0.003	0.025	0.010	0.013	0.006	0.013	0.010	0.105	0.012	0.074	0.009	0.013	0.023	0.009
15	Dannon Light N Fit 0.375	0.050	0.013	0.019	0.010	0.013	0.006	0.012	0.010	0.136	0.018	0.103	0.011	0.014	0.030	0.010
16	Dannon Light N Fit 1.5	-2.714	0.016	0.015	0.010	0.013	0.007	0.012	0.010	0.158	0.017	0.120	0.014	0.014	0.031	0.010
17	Dannon Nutriday 1.0	0.043	-2.094	0.007	0.010	0.021	0.002	0.025	0.016	0.205	0.030	0.166	0.011	0.019	0.055	0.017
18	Stonyfield Farm 0.375	0.045	0.009	-4.062	0.011	0.017	0.008	0.015	0.011	0.129	0.016	0.093	0.012	0.014	0.027	0.010
19	Stonyfield Farm 1.5	0.050	0.003	0.038	-4.143	0.012	0.005	0.013	0.010	0.093	0.015	0.066	0.008	0.015	0.026	0.009
20	Yoplait 0.375	0.044	0.011	0.014	0.010	-4.351	0.008	0.013	0.009	0.127	0.013	0.094	0.011	0.011	0.024	0.009
21	Yoplait 1.0	0.047	0.000	0.008	0.005	0.013	-4.458	0.010	0.008	0.131	0.012	0.090	0.010	0.008	0.021	0.005
22	Yoplait Delights 1.0	0.038	0.011	0.017	0.011	0.018	0.009	-4.365	0.010	0.136	0.015	0.102	0.011	0.012	0.024	0.011
23	Yoplait Fiber One 1.0	0.041	0.010	0.019	0.011	0.018	0.008	0.016	-4.115	0.145	0.018	0.105	0.012	0.014	0.028	0.011
24	Yoplait Light 0.375	0.055	0.016	0.016	0.010	0.014	0.007	0.013	0.010	-3.112	0.017	0.121	0.013	0.013	0.031	0.010
25	Yoplait Light Thick & Creamy 0.375	0.053	0.016	0.016	0.009	0.015	0.007	0.013	0.010	0.154	-3.231	0.119	0.013	0.015	0.031	0.010
26	Yoplait Original 0.375	0.055	0.016	0.016	0.010	0.014	0.007	0.013	0.010	0.153	0.017	-3.156	0.013	0.013	0.031	0.010
27	Yoplait Original 1.5	0.062	0.005	0.013	0.008	0.015	0.007	0.011	0.009	0.159	0.014	0.117	-3.092	0.013	0.028	0.008
28	Yoplait Thick & Creamy 0.375	0.057	0.016	0.016	0.009	0.014	0.007	0.013	0.010	0.155	0.018	0.114	0.014	-3.231	0.030	0.010
29	Yoplait Trix 1.5	0.050	0.016	0.018	0.011	0.018	0.009	0.014	0.010	0.158	0.019	0.125	0.014	0.015	-3.276	0.011
30	Yoplait Yo Plus 1.0	0.035	0.010	0.019	0.010	0.018	0.007	0.017	0.012	0.156	0.020	0.124	0.011	0.015	0.027	-3.890

Table A.7: Estimating the Revenue Sharing Parameter: Marginal Cost Approach, Excluding Chobani

	(1) Full sample	(2) Excluding Q3/Q4 2011
Yoplait * Pre July 2011	0.0353	0.0507
	[0.0349, 0.0357]	[0.0501, 0.0512]
	2,956,461	2,459,352
Implied Rev. Sharing Coef.	0.9653	0.9506
	[0.9650, 0.9657]	[0.9501, 0.9511]

Notes: Bootstrapped 95-percent confidence intervals in brackets. An observation is a product–store–week combination. The sample excludes Chobani products. The estimates are based on the demand system in Table 2 (column 4). In all specifications we reject the hypothesis that $\hat{\mu}$ is equal to 1.

Table A.8: Estimating the Revenue Sharing Parameter: Indirect Inference Approach

	Full sample		Excl. 2nd half 2011	
Rev. sharing coef.	Estimate	95% CI	Estimate	95% CI
0.940	0.0187	[0.0184, 0.0189]	0.0067	[0.0064, 0.0069]
0.945	0.0156	[0.0153, 0.0158]	0.0036	[0.0034, 0.0038]
0.950	0.0125	[0.0122, 0.0127]	0.0005	[0.0003, 0.0007]
0.955	0.0093	[0.0092, 0.0095]	-0.0026	[-0.0028, -0.0024]
0.960	0.0063	[0.0061, 0.0064]	-0.0056	[-0.0058, -0.0055]
0.965	0.0032	[0.0030, 0.0033]	-0.0087	[-0.0089, -0.0086]
0.970	0.0001	[-0.0001, 0.0002]	-0.0118	[-0.0119, -0.0116]
0.975	-0.0030	[-0.0031, -0.0029]	-0.0148	[-0.0149, -0.0147]
0.980	-0.0061	[-0.0062, -0.0060]	-0.0179	[-0.0180, -0.0178]
0.985	-0.0092	[-0.0092, -0.0091]	-0.0209	[-0.0210, -0.0209]
0.990	-0.0122	[-0.0123, -0.0122]	-0.0240	[-0.0240, -0.0239]
0.995	-0.0153	[-0.0153, -0.0153]	-0.0270	[-0.0270, -0.0270]
1.000	-0.0184	[-0.0184, -0.0184]	-0.0301	[-0.0301, -0.0301]

Notes: Coefficients on the post-July 2011 * Yoplait indicator from regressions of log counterfactual prices on a post-July 2011 * Yoplait indicator, display and feature indicators, product-store, and week-store fixed effects. Confidence intervals computed from the 2.5th and 97.5th percentiles of 200 bootstrap replications.

Table A.9: The Impact of an Efficient Vertical Contract on Market Outcomes: The Role of Competition

	(1)	(2)	(3)	(4)	(5)	(6)
	Price change (in log points)		Market share change (in log points)		Profit change (in log points)	
	BR	Equilibrium	BR	Equilibrium	BR	Equilibrium
Panel A: Revenue sharing parameter 0.97						
Chobani		0.00016	-0.00549	-0.00627	0.13950	0.13942
		(0.00000)	(0.00005)	(0.00006)	(0.00066)	(0.00066)
General Mills	-0.01881	-0.01881	0.06192	0.06185	0.09408	0.09400
	(0.00006)	(0.00006)	(0.00034)	(0.00034)	(0.00029)	(0.00029)
Groupe Danone		-0.00016	-0.00716	-0.00675	0.09496	0.09496
		(0.00000)	(0.00005)	(0.00005)	(0.00030)	(0.00030)
Observations	1,811,560	1,811,560	1,811,560	1,811,560	1,811,560	1,811,560
Panel B: Revenue sharing parameter 0.95						
Chobani		0.00027	-0.00927	-0.01060	0.24592	0.24579
		(0.00000)	(0.00009)	(0.00010)	(0.00124)	(0.00124)
General Mills	-0.03145	-0.03145	0.10293	0.10280	0.15736	0.15723
	(0.00010)	(0.00010)	(0.00057)	(0.00057)	(0.00048)	(0.00048)
Groupe Danone		-0.00026	-0.01210	-0.01143	0.16477	0.16478
		(0.00001)	(0.00009)	(0.00009)	(0.00054)	(0.00054)
Observations	1,811,560	1,811,560	1,811,560	1,811,560	1,811,560	1,811,560

Notes: The estimates are based on demand estimates in Table 2 (column 4). Standard errors in parentheses. An observation is a product–store–week combination. We restrict attention to weeks prior to July 2011 (i.e., the time when the transaction was completed). Each column displays regression coefficients of $\log(X^{\text{counterfactual}}) - \log(X^{\text{observed}})$ on firm-level indicators, for $X \in \{\text{price}, \text{market share}, \text{profit}\}$. $X^{\text{counterfactual}}$ are the equilibrium outcomes with an efficient vertical contract in columns labeled 'Equilibrium'; whereas in columns labeled 'BR', $X^{\text{counterfactual}}$ is the best response of General Mills to rivals' prices when an efficient vertical contract is implemented and rivals are not allowed to respond. X^{observed} is the equilibrium outcome in the observed equilibrium.

B A Supply-Side Model with a Retail Sector

We assume the existence of manufacturers and a retailer. Manufacturers simultaneously set the wholesale prices of their portfolio of products to maximize their profits at the market level. The retailer takes wholesale prices as given and chooses retail prices according to a pricing equation discussed below.

Specifically, we assume that the manufacturing firm f (e.g., General Mills), selling product portfolio $\mathcal{J}_{fst}$ at store s in week t, solves

$$\max_{w_{jst} \forall j \in \mathcal{J}_{fst}} \sum_{j \in \mathcal{J}_{fst}} (w_{jst}\mu_{f(j)t} - c_{jst}) M_{st} s_{jst}(\mathbf{p}(\mathbf{w})),$$

where $\mu_{f(j)t}$ corresponds to the share of revenues that firm f keeps (and $1-\mu_{f(j)t}$ corresponds to the share of revenues that f pays to the upstream firm), c_{jst} is the manufacturer's marginal cost of product j at store s in week t, $\mathbf{p}(\mathbf{w})$ is the retail price correspondence given the vector of wholesale prices $\mathbf{w}$, and M_{st} represents the size of the market.

The first-order conditions for the manufacturing firm f are given by

$$s_{jst}(\mathbf{p}(\mathbf{w})) + \sum_{k \in \mathcal{J}_{fst}} \sum_{h \in \mathcal{J}_{fst}} (w_{kst} - c_{kst}/\mu_{f(k)t}) \frac{\partial s_{kst}(\mathbf{p}(\mathbf{w}))}{\partial p_{hst}} \frac{\partial p_{hst}}{\partial w_{jst}} = 0, \tag{7}$$

where we use that $\mu_{f(k)t} = \mu_{f(j)t}$ for all $j, k \in \mathcal{J}_{fst}$.

We follow Miller and Weinberg (2017) and assume that the retail prices are determined by the system of equations

$$0 = \lambda s_{jst} + \sum_{k \in J_{st}} \frac{\partial s_{kst}(\mathbf{p})}{\partial p_{jst}} (p_{kst} - w_{kst}), \quad \forall j \in J_{st}, \tag{8}$$

where $\lambda \in [0,1]$ and $J_{st} = \bigcup_f J_{fst}$. This system of equations is identical to the system of first-order equations of a retailer firm that maximizes its profit, except for the presence of the scaling parameter λ. The parameter λ scales the retail markups between zero ($\lambda = 0$) and the monopoly markups ($\lambda = 1$), capturing in a simplified way the competitive pressure faced by the retailer. We assume that the retailer's marginal cost of selling a good equals the wholesale price of that good.

Combining Equation 7 and Equation 8 yields an equation like Equation 4, where we can recover the ratios $c_{kst}/\mu_{f(k)t}$ for every product–store–week combination, as a function of prices, demand estimates, and a value of λ (the scaling parameter in the retail prices equations).

C Contracts

In this Appendix, we provide documentation regarding the contract between General Mills and Sodima/Sodiaal.

A copy of the original contract between the two can be found in `https://www.sec.gov/Archives/edgar/data/40704/000089710107001522/gen072744s1_ex10-32.htm`. This contract establishes the obligations and rights of the parties (e.g., manufacturing, property, machine maintenance, assistance) and the payment structure.

By 2010, this contract had been amended twelve times. Examples include

- Historical record: `https://www.sec.gov/Archives/edgar/data/40704/000089710107001522/gen072744s1_ex10-32.htm`. Includes information about:
  - 1st amendment (1977)
  - 2nd amendment (1981)
  - 3rd amendment (undated in the public record)
  - 5th amendment (undated in the public record)
  - 7th amendment (1999)
  - 8th amendment (2002)
- 9th amendment: `https://www.sec.gov/Archives/edgar/data/40704/000089710108000662/genmills081060_ex10-1.htm`
- 10th amendment: `https://www.sec.gov/Archives/edgar/data/40704/000095013709002011/c50087exv10w17.htm`
- 11th amendment: `https://www.sec.gov/Archives/edgar/data/40704/000095012309071898/c55115exv10w2.htm`
- 12th amendment: `https://www.sec.gov/Archives/edgar/data/40704/000095012310114560/c61873exv10w4.htm`

We conclude that contract amendments, though not occurring every year, were not uncommon and, to the best of our knowledge, did not result in termination threats.

D The impact of the Yoplait acquisition on General Mills' day-to-day stock returns

Two facts suggest the transaction likely added value to the company. On the one hand, General Mills' stock price increased by 1.44 percent following the announcement of the deal reached with PAI Partners.[32] On the other hand, given the transaction fee of $1.1 billion, the profit gains mentioned above imply a rate of return of the order of 5.6 to 9.5 percent, which are of the same order as the reciprocal of the P/E ratio of General Mills at the time of the transaction (e.g., its P/E ratio in 2011 was in the 14-15 range).

We explicitly consider the impact of the transaction on the stock prices of General Mills and Danone (Chobani is privately owned so we cannot perform the analyses that follow). We use the daily returns to test whether the returns that followed relevant dates (such as the announcement of the deal as well as when it was closed) were significantly different from those experienced in the two years before the transaction. We did this considering both the day-to-day returns and considering where the returns rank in the corresponding distributions. We find that though daily returns were positive for both General Mills (1.44%) and Danone (1.08%), in both cases we reject that these are abnormal or that these rank in the top 5 percent of returns during our sample period.

We also conducted an in-depth search of media articles from the time period of interest focusing on General Mills. We focused on identifying articles that explicitly mentioned stock price changes following the different announcements. We were able to find the following:

- General Mills filed for arbitration following Sodima's announcement of its intention to terminate the agreement. General Mills filed after the closing of the stock market and the source writes "In after-hours trading, General Mills stock was up a few pennies."[33]
- General Mills' stock price increased by 1.44 percent following the announcement of the deal reached with PAI Partners.[34]
- General Mills' stock price increased by 2 cents (0.05 percent) after it was announced that the deal with PAI Partners and Sodiaal had received regulatory approval.[35]

[32] Mike Hughlett, "General Mills confirms deal to buy 50% of Yoplait for $1.1B", *StarTribune*, March 19th, 2011. Accessed through the Texas A&M University Libraries on August 20th, 2025.

[33] Mike Hughlett, "No more Yoplait? General Mills says no way", *StarTribune*, September 11th, 2010. Accessed through the Texas A&M University Libraries on August 20th, 2025.

[34] Mike Hughlett, "General Mills confirms deal to buy 50% of Yoplait for $1.1B", *StarTribune*, March 19th, 2011. Accessed through the Texas A&M University Libraries on August 20th, 2025.

[35] Business Briefs, "General Mills to control Yoplait", *Investor's Business Daily*, May 19th, 2011; Mike Hughlett "General Mills buying biggest bite of Yoplait", *StarTribune*, May 19th, 2011; Accessed through the Texas A&M University Libraries on August 20th, 2025.

E Examining the determinants of the supply chain's profit-maximizing μ

The model that we presented in section 2 showed that the derivative of the supply chain's profits with respect to the revenue-sharing parameter μ can be positive or negative, meaning that such a contract may increase supply chain profits despite causing inefficiency. In this Appendix, we show that this result is not specific to the Hotelling-style model presented in section 2; it holds in other settings as well.

Specifically, we consider a nested logit in which all inside (single-product) firms share a single nest, and the outside option is alone in its own nest, with within-nest correlation ρ. Conditional share and nest share are

$$s_{j|g} = \frac{\exp((-\alpha p_j + \xi_j)/(1-\rho))}{\sum_{k\in g} \exp((-\alpha p_k + \xi_k)/(1-\rho))}, \qquad s_j = s_{j|g}\, s_g,$$

so that $\rho = 0$ recovers the plain logit and $\rho \to 1$ makes the inside firms perfect substitutes. Firm 2 is subject to a revenue-sharing agreement with parameter μ. We also allow for additional non-contracted rivals, while firm 2 remains the only firm subject to the agreement.

In this model, μ, α, ρ, marginal costs, and ξ determine equilibrium prices and profits. In what follows, we focus on symmetric cases (i.e., a common ξ and common marginal costs of 1) because this symmetry simplifies the explanation of our findings; however, our findings do not depend on this symmetry.

We focus on the supply chain's profits, $\Pi(\mu) = (p_2(\mu) - c)\ s_2(p(\mu))$, and on the value μ^* that maximizes them. Lowering μ has two opposing effects: a *direct* cost, because firm 2 prices inefficiently from the chain's standpoint, and a *strategic* benefit, because rivals facing a softer competitor raise their own prices. Whether an interior optimum $\mu^* < 1$ arises depends on which effect dominates.

Figure A.7 plots μ^* against the within-nest correlation ρ, one line per number of non-contracted rivals, holding the price coefficient fixed at $\alpha = 1$. Initially, as ρ rises, the products become closer substitutes, the strategic channel strengthens, and μ^* first falls below one, but this reverts at higher values of ρ, with μ^* increasing towards one again (this happens even in the 1-rival example, for which the line only reaches its minimum near the right edge, so its recovery toward one lies just beyond the plotted range.). Finally, for a fixed ρ, adding non-contracted rivals pushes μ^* back toward one, as demand becomes more elastic with a larger number of firms.

Although this is a stylized example that could be generalized further, we believe it illustrates well that, in general, the optimal μ can be interior or on the boundary, depending on demand elasticity, product substitutability, and the number of products in the market, among others.

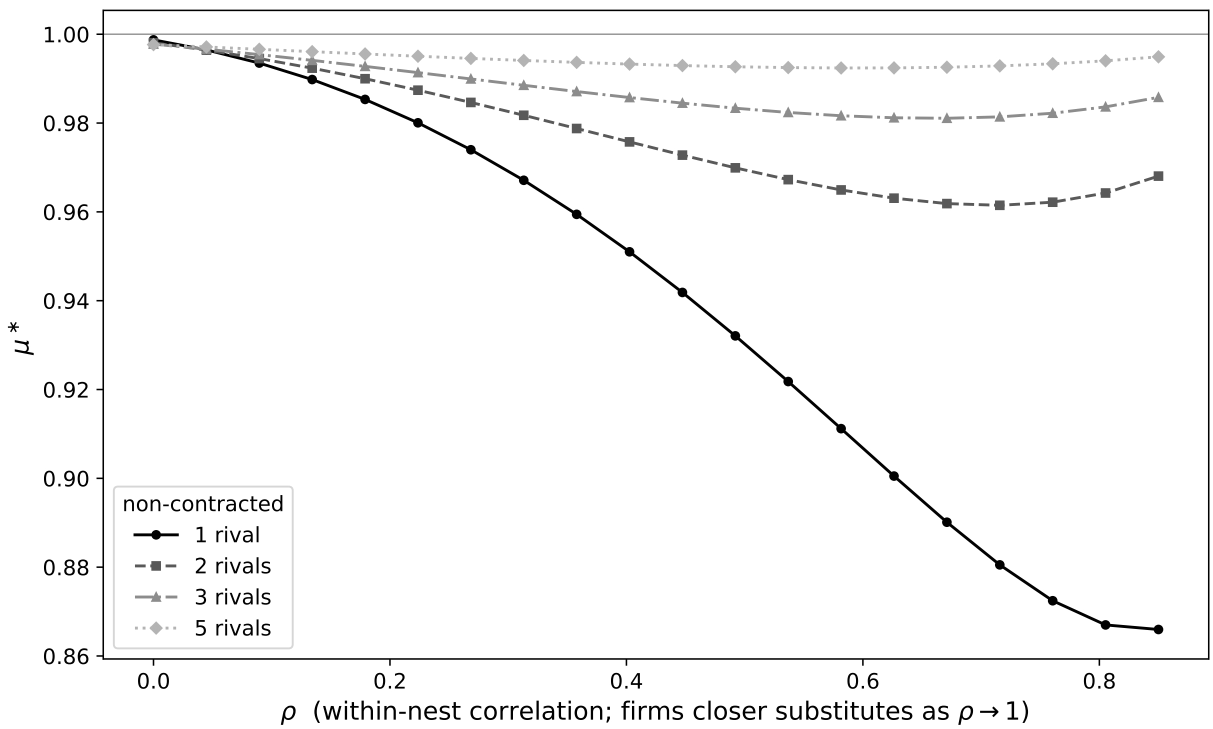


Figure A.7: Nested logit: chain-optimal μ^* versus the within-nest correlation ρ, by number of non-contracted rivals.